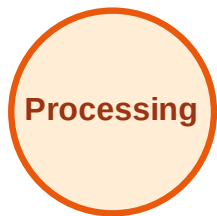


BFS Traversal

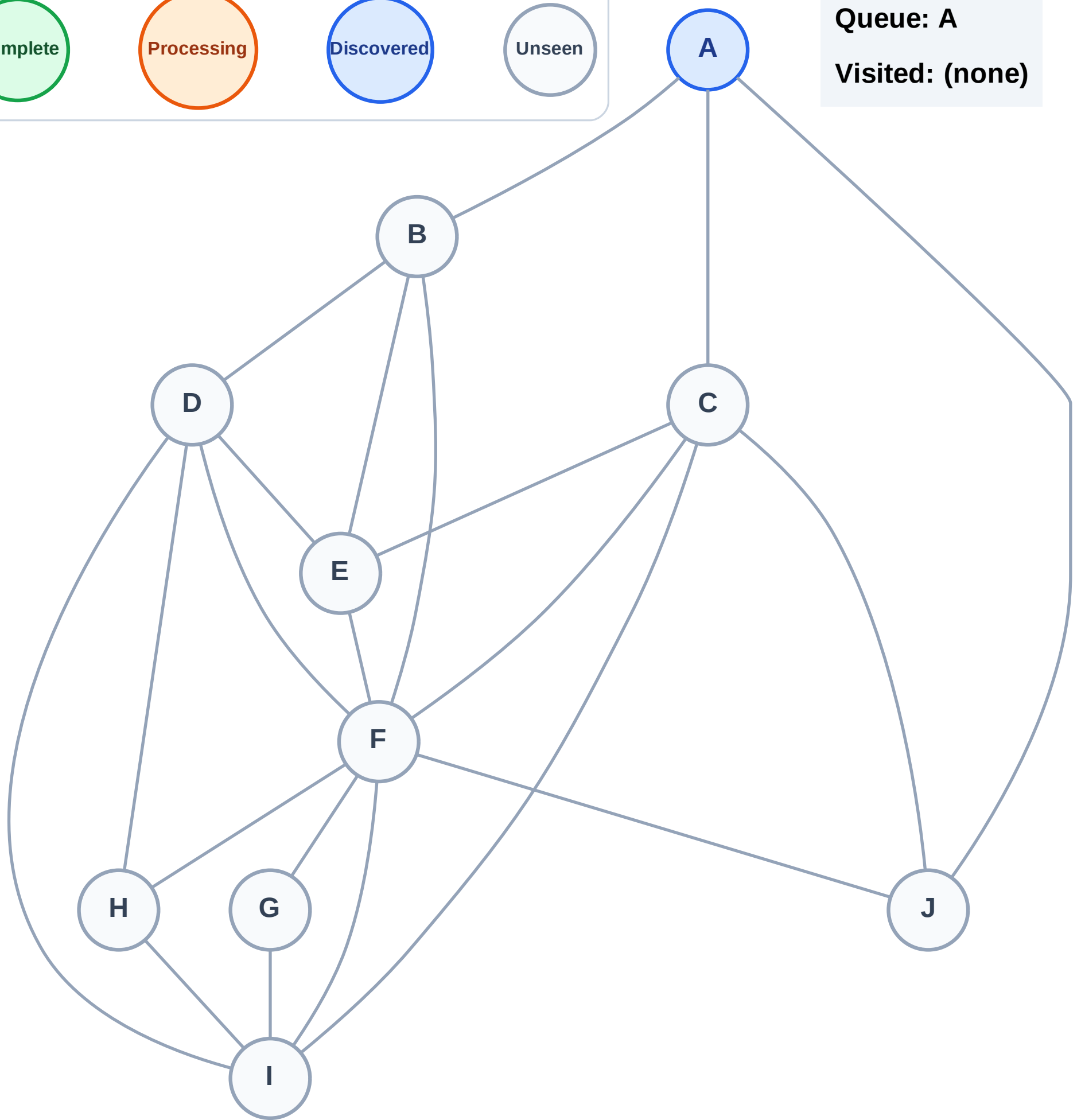
Step 1: Start at A

Legend



Queue: A

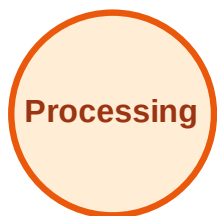
Visited: (none)



BFS Traversal

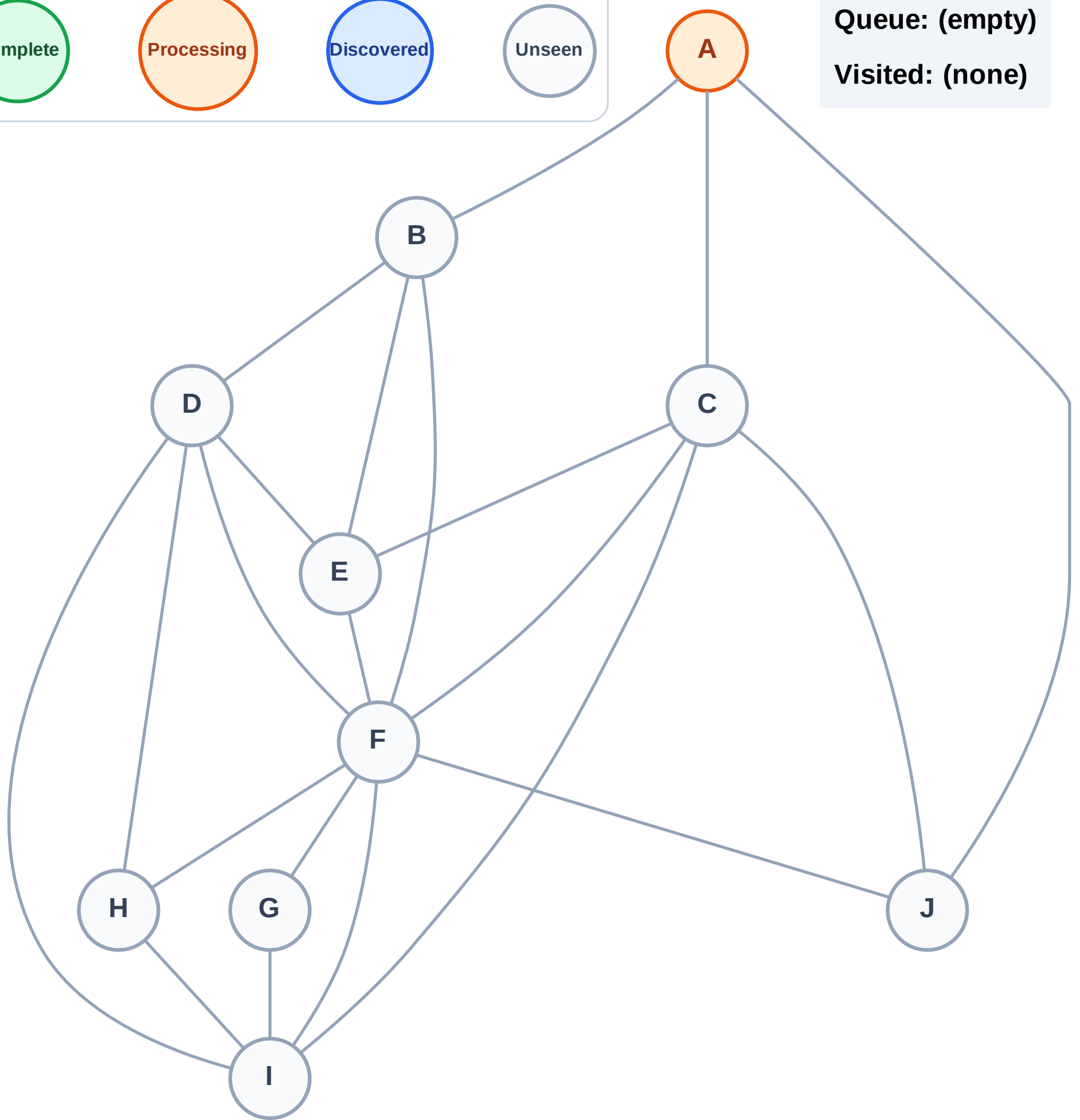
Step 2: Process node A

Legend



Queue: (empty)

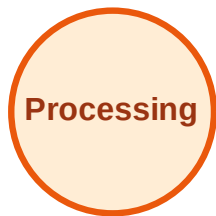
Visited: (none)



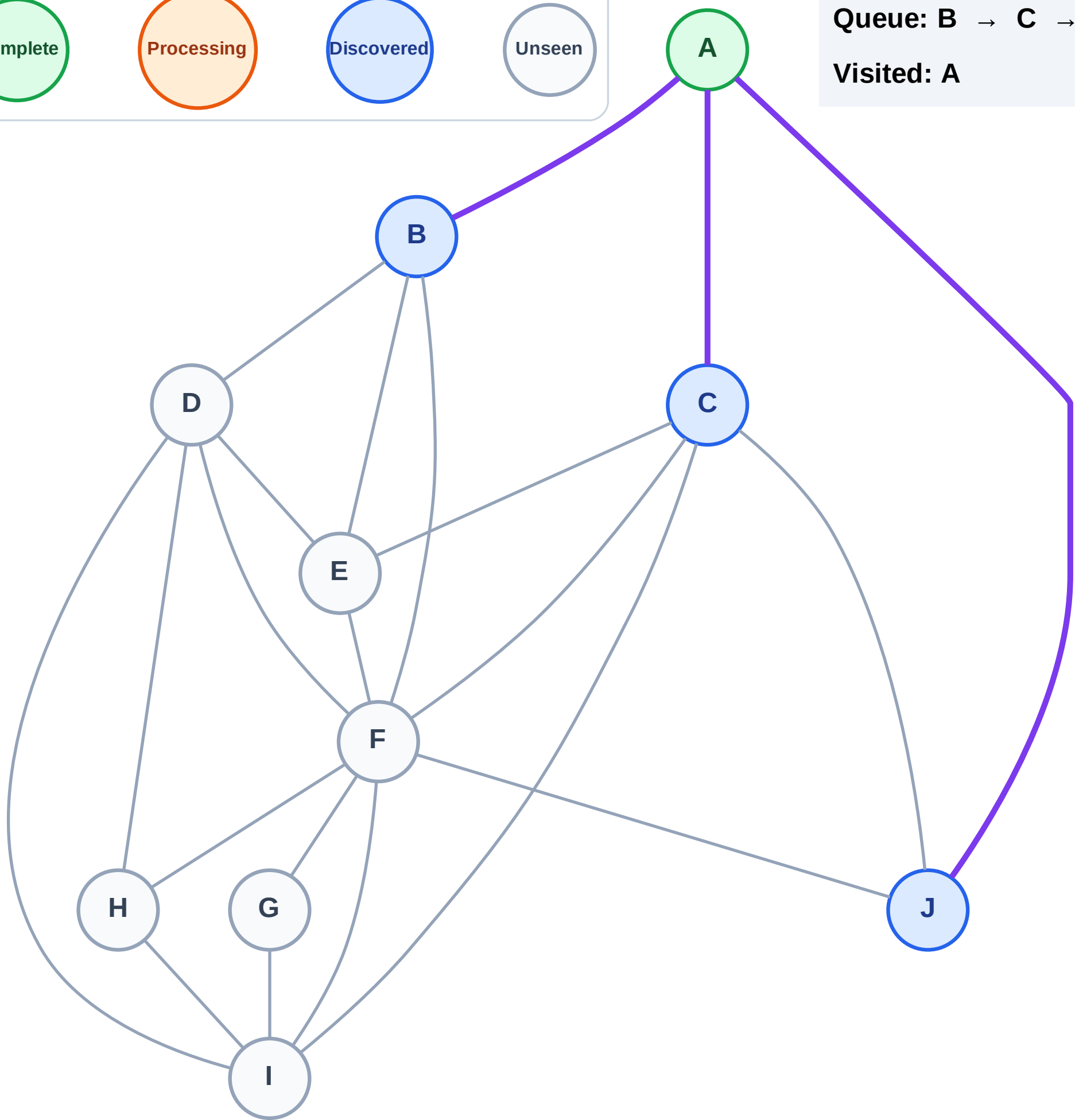
BFS Traversal

Step 3: Discover B, C, J from A

Legend



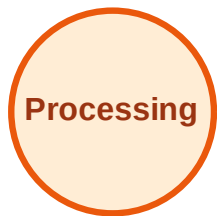
Queue: B → C → J
Visited: A



BFS Traversal

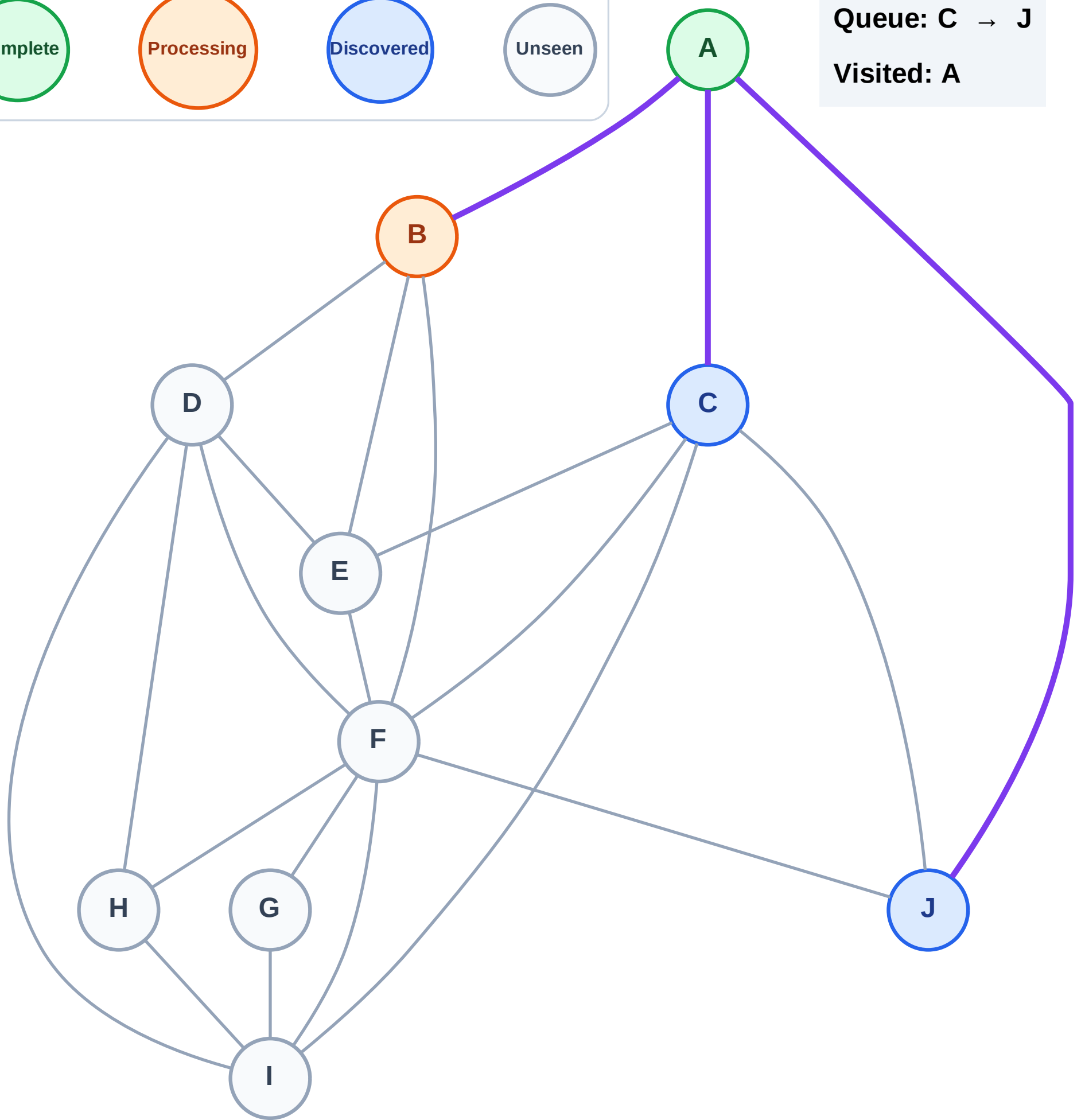
Step 4: Process node B

Legend



Queue: C → J

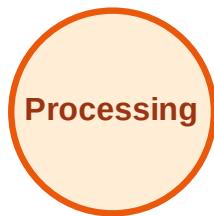
Visited: A



BFS Traversal

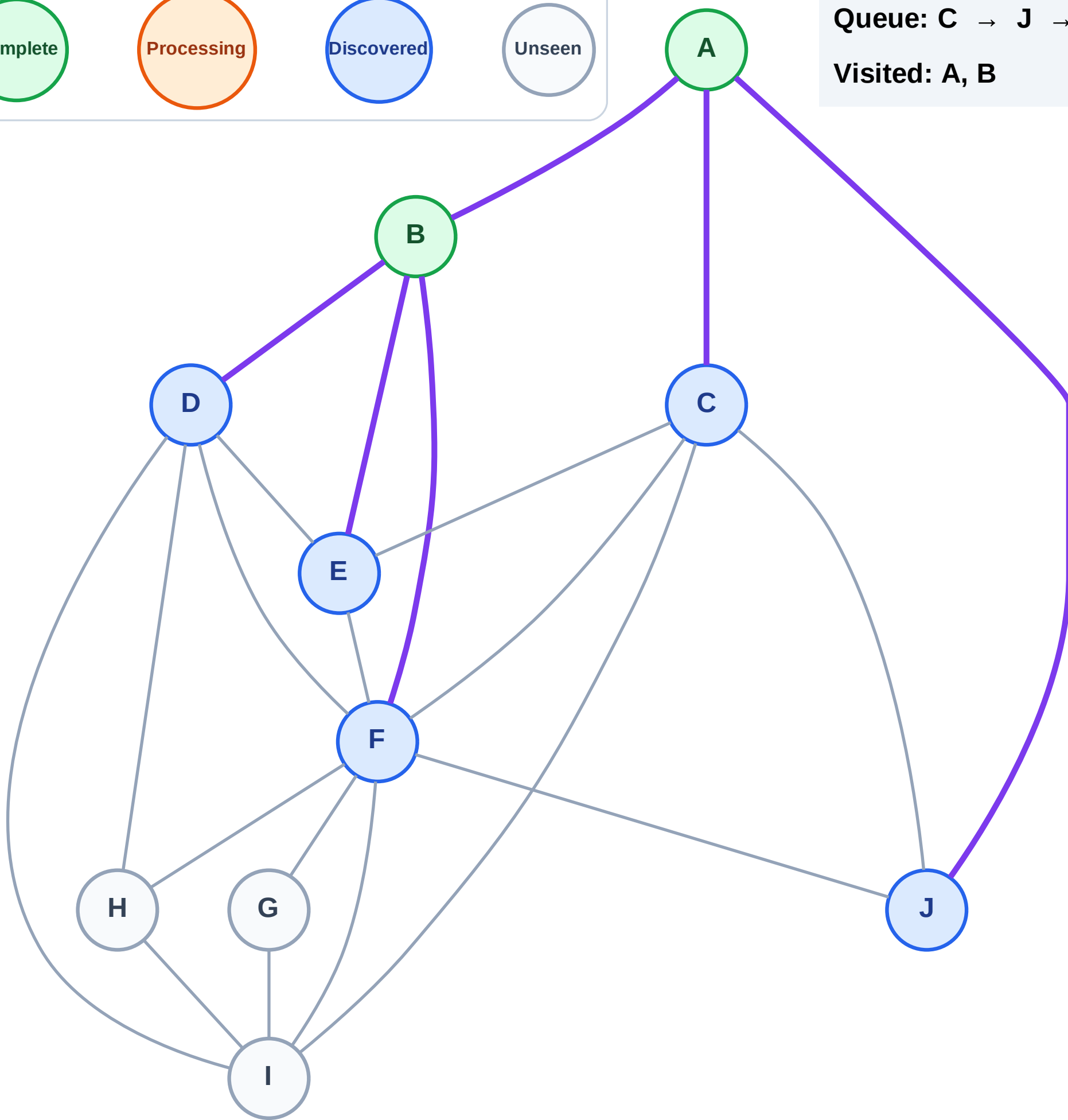
Step 5: Discover D, E, F from B

Legend



Queue: C → J → D → E → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

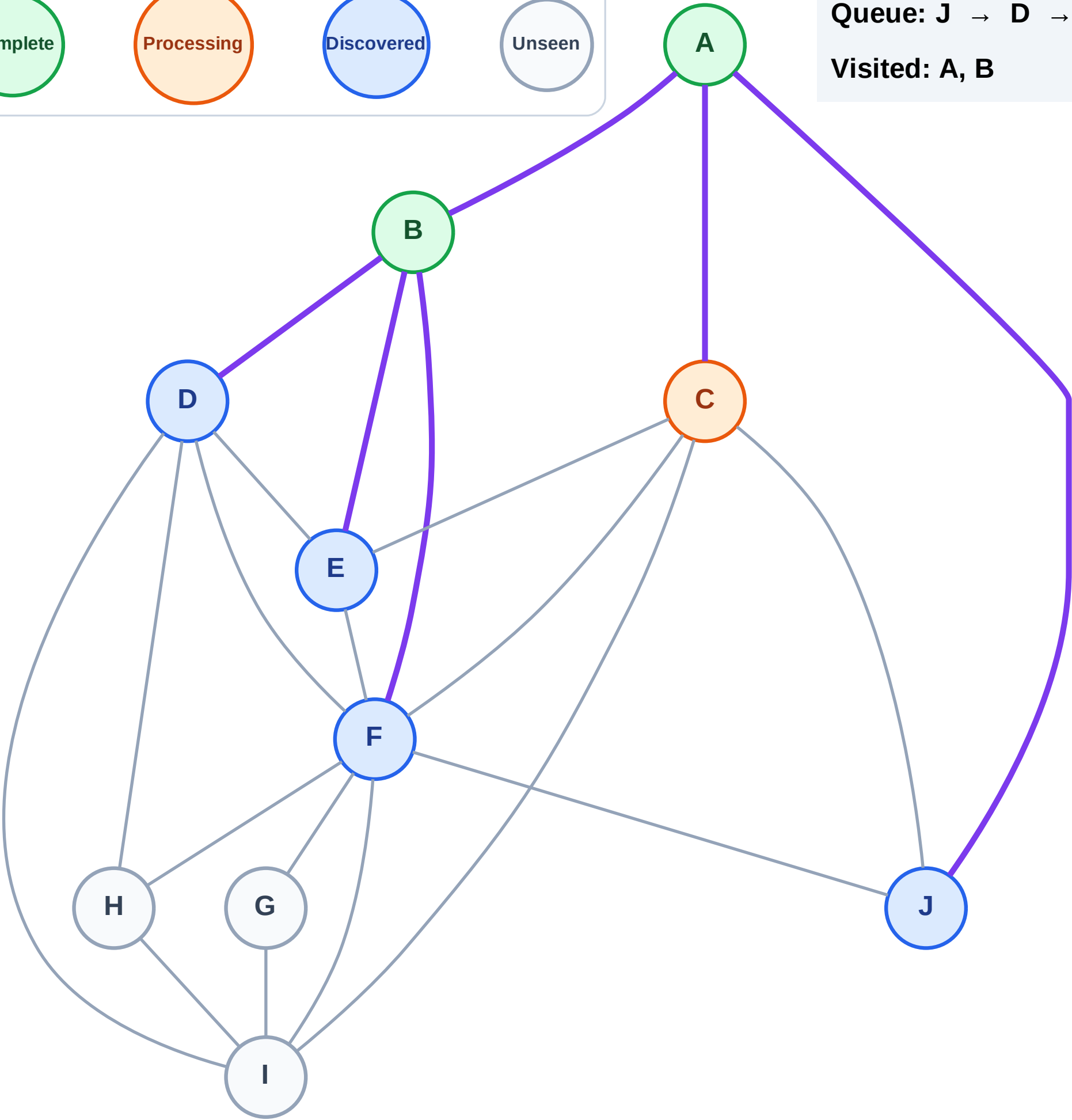
Processing

Discovered

Unseen

Queue: J → D → E → F

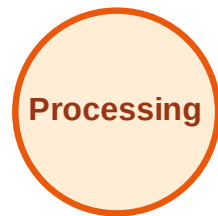
Visited: A, B



BFS Traversal

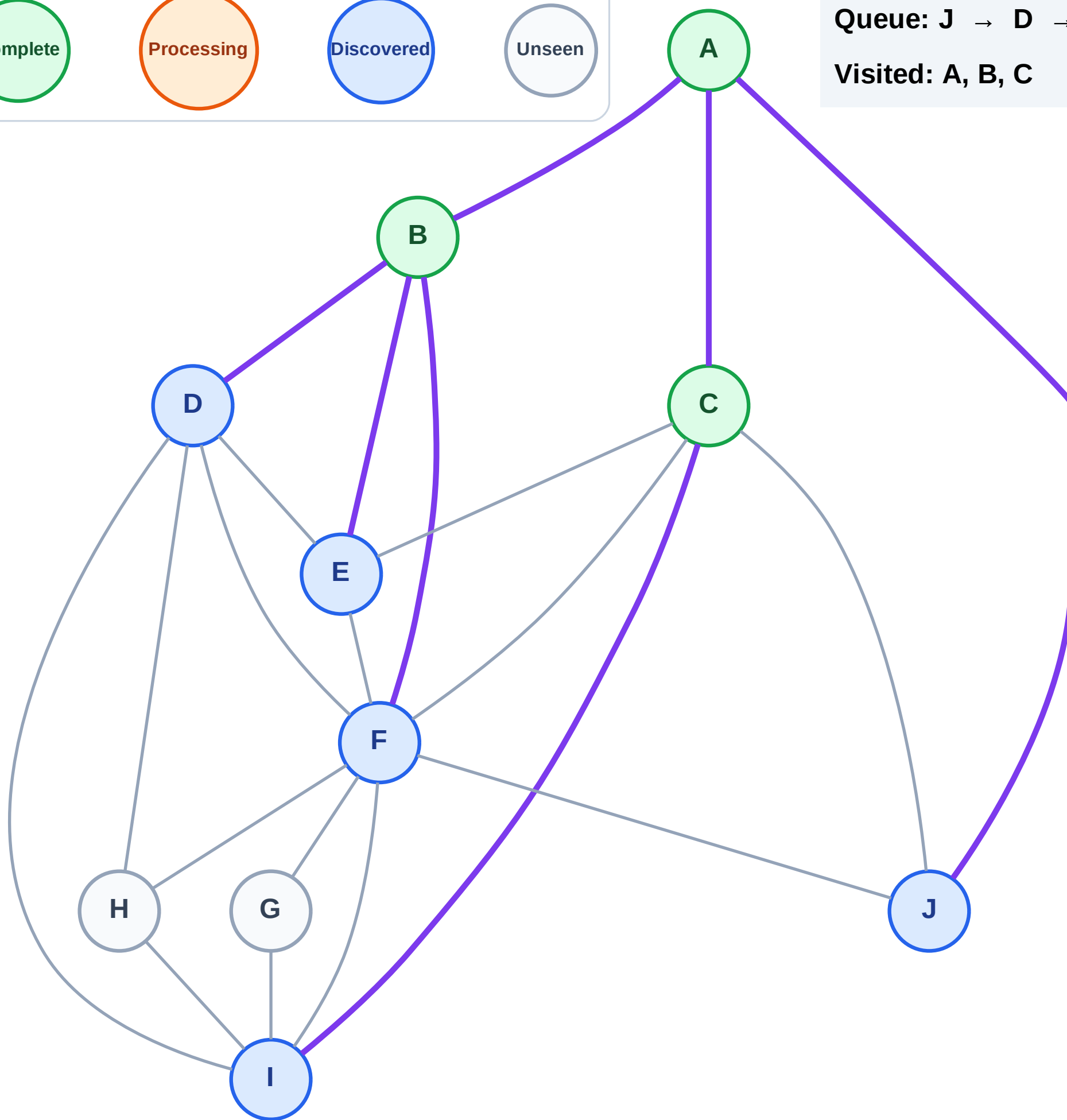
Step 7: Discover I from C

Legend



Queue: J → D → E → F → I

Visited: A, B, C



BFS Traversal

Step 8: Process node J

Legend

Complete

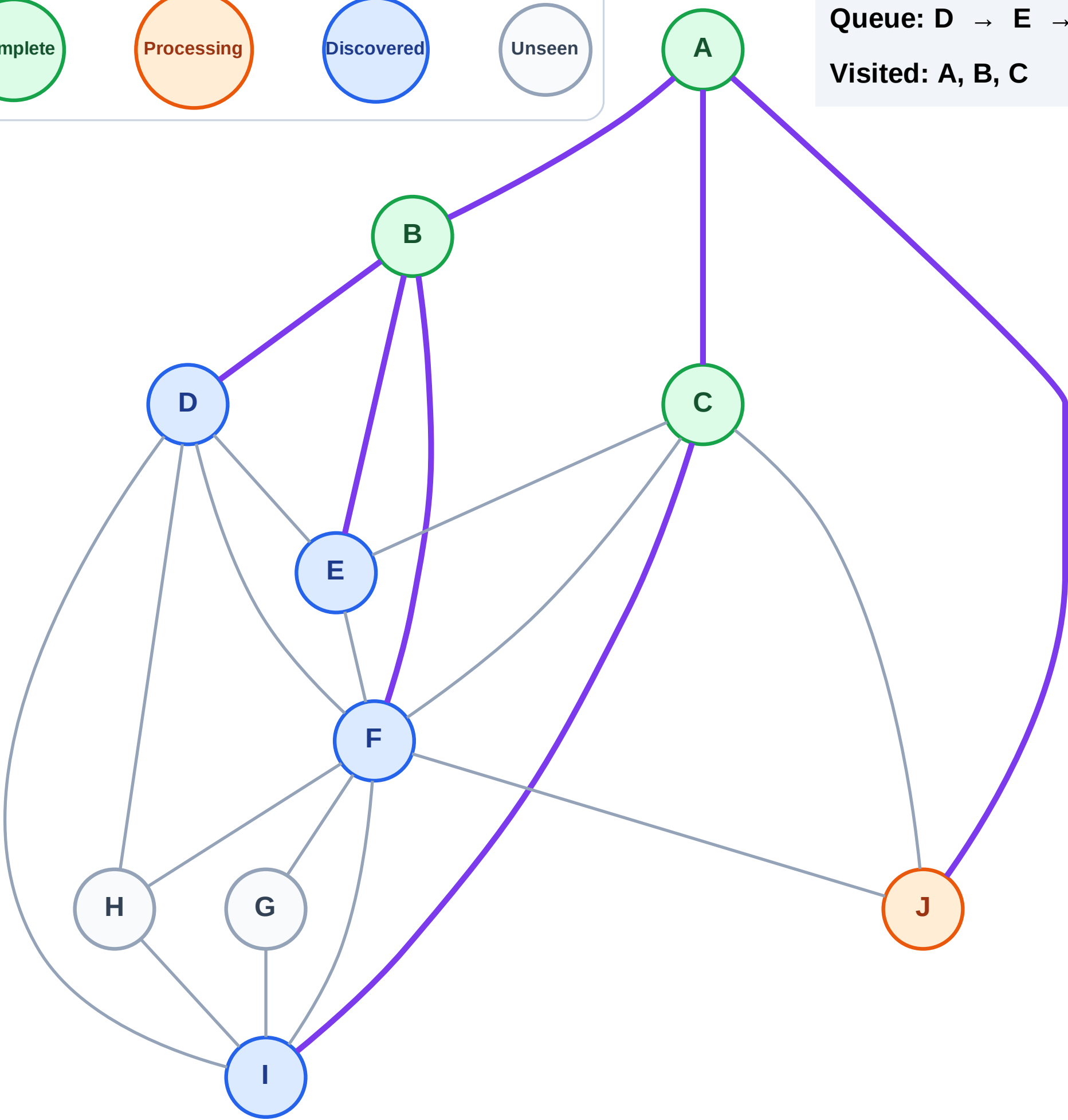
Processing

Discovered

Unseen

Queue: D → E → F → I

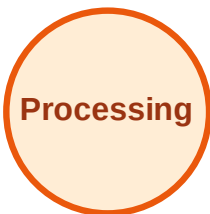
Visited: A, B, C



BFS Traversal

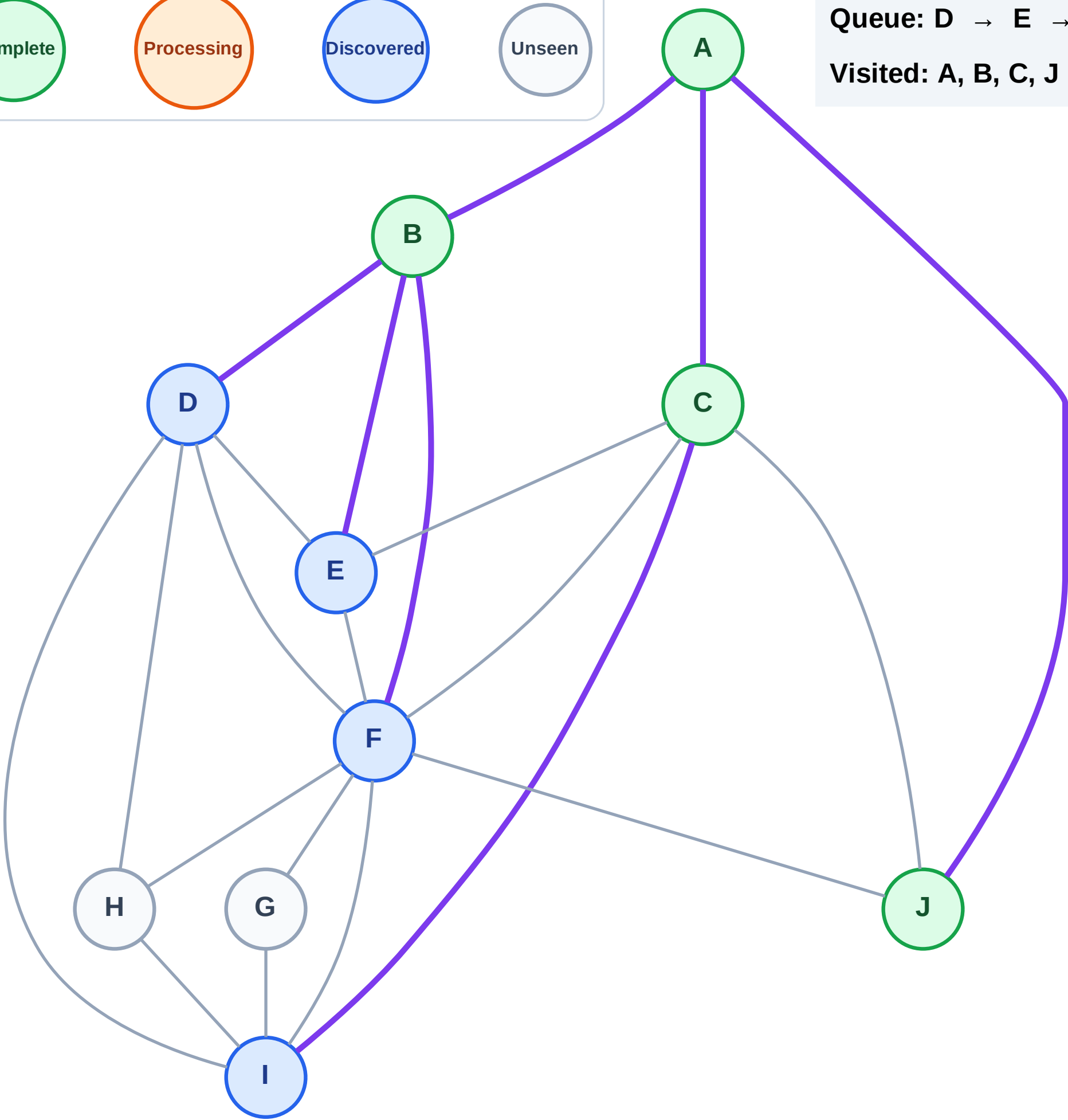
Step 9: No new neighbors from J

Legend



Queue: D → E → F → I

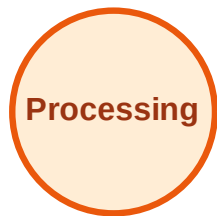
Visited: A, B, C, J



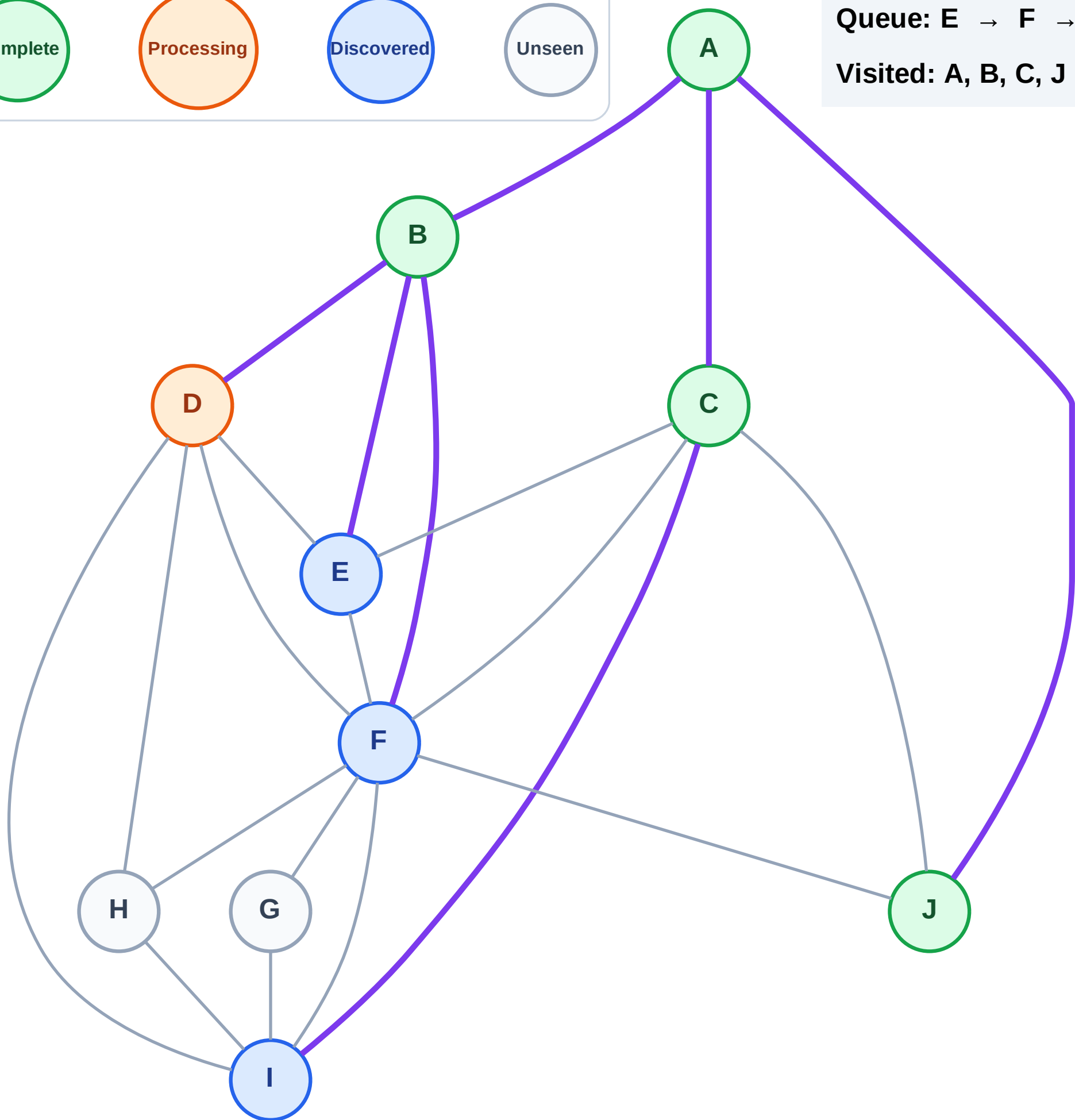
BFS Traversal

Step 10: Process node D

Legend



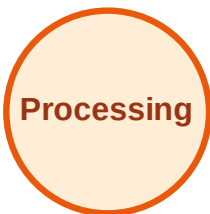
Queue: E → F → I
Visited: A, B, C, J



BFS Traversal

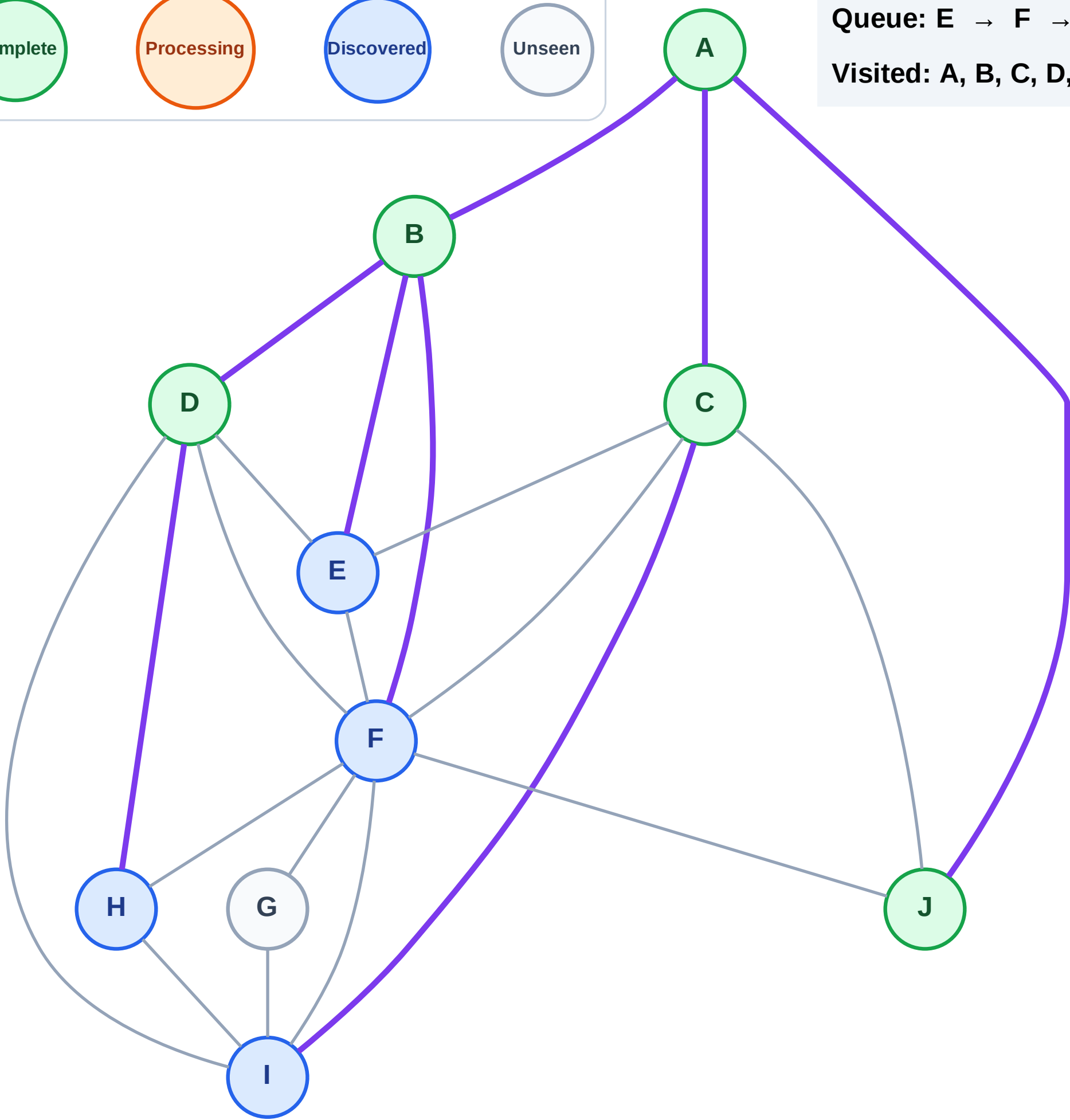
Step 11: Discover H from D

Legend



Queue: E → F → I → H

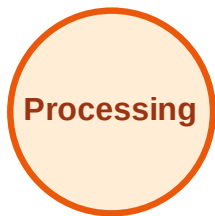
Visited: A, B, C, D, J



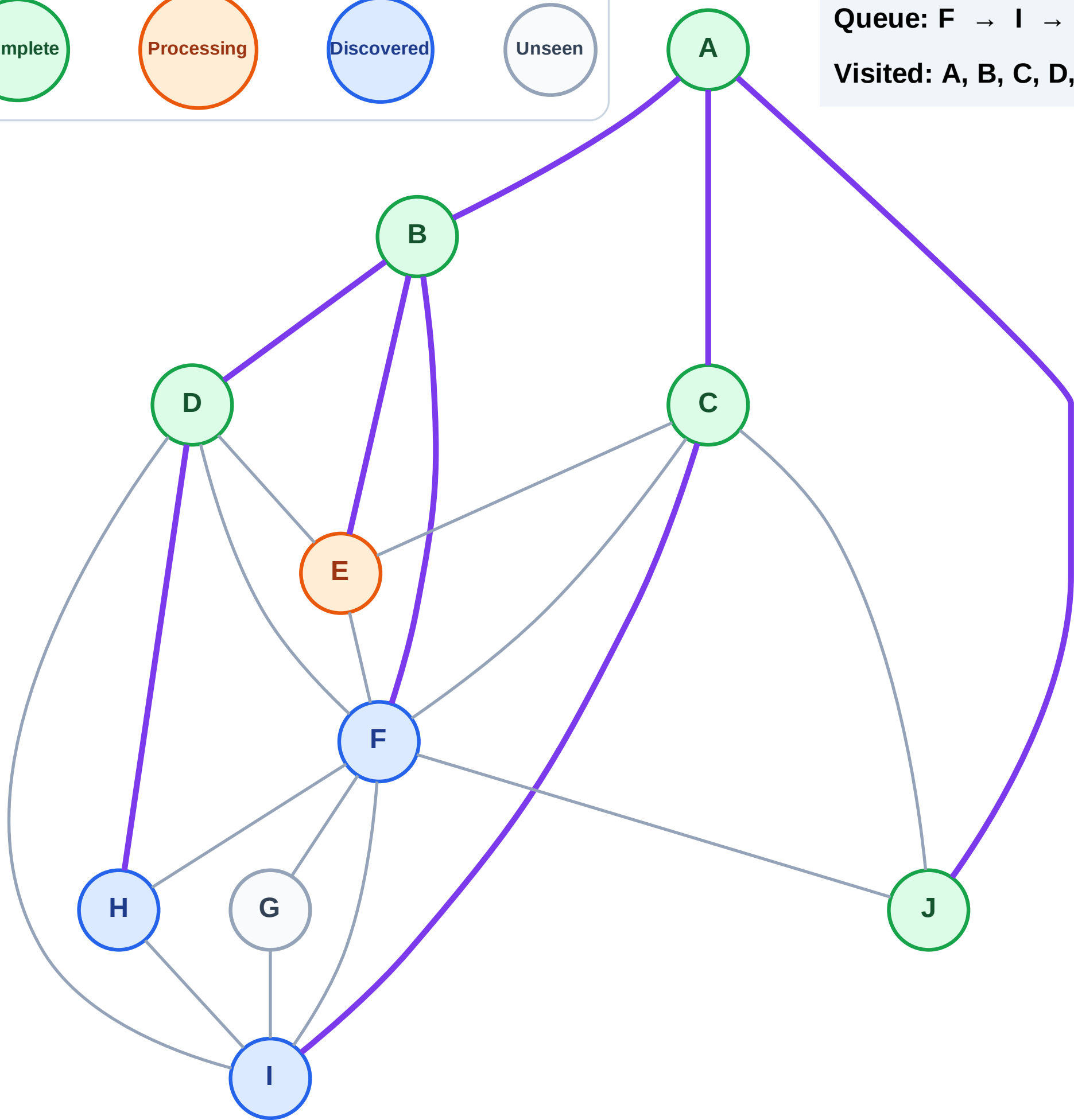
BFS Traversal

Step 12: Process node E

Legend



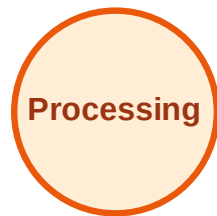
Queue: F → I → H
Visited: A, B, C, D, J



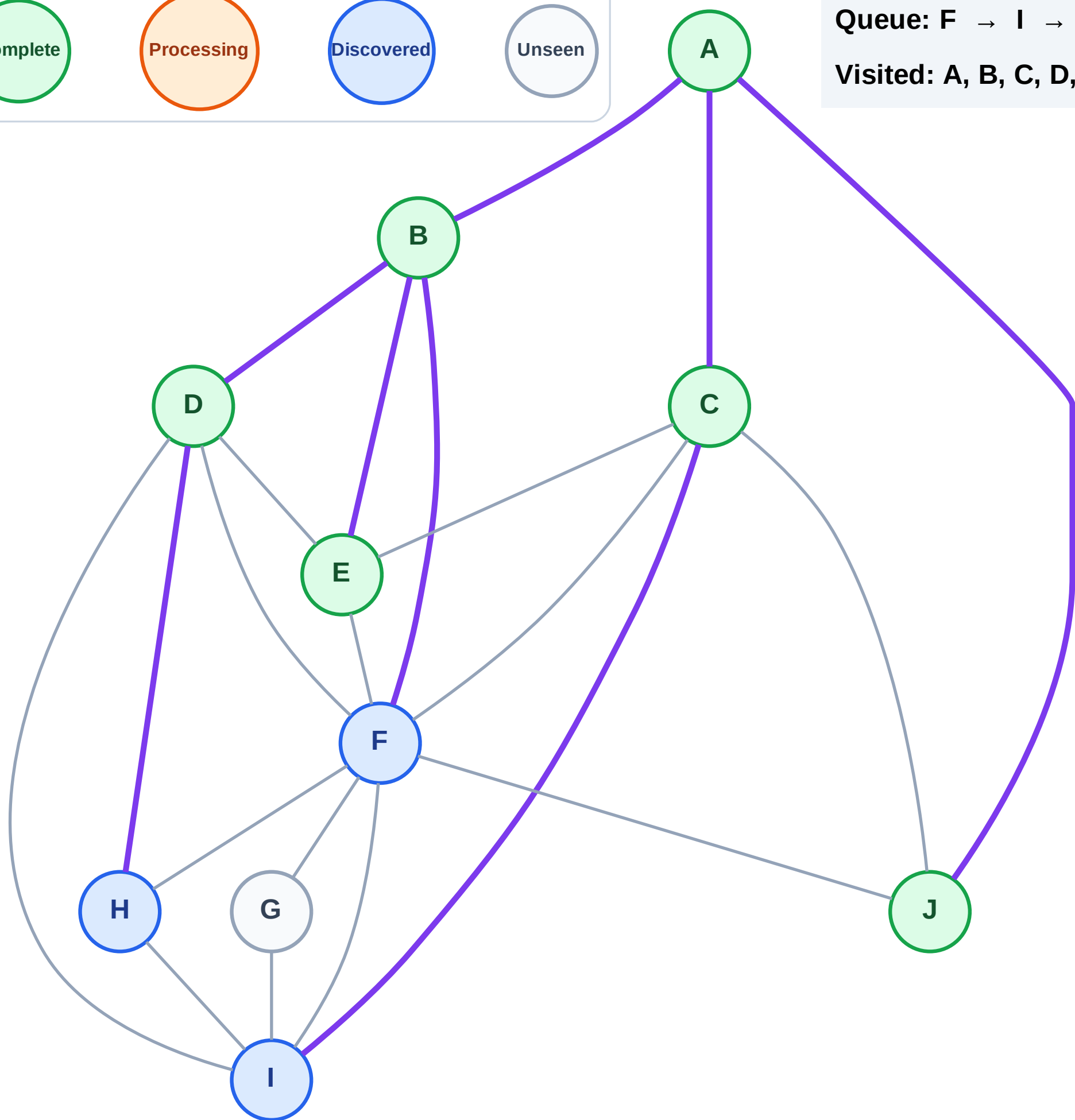
BFS Traversal

Step 13: No new neighbors from E

Legend



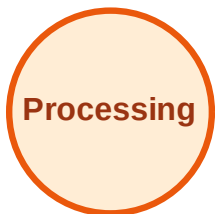
Queue: F → I → H
Visited: A, B, C, D, E, J



BFS Traversal

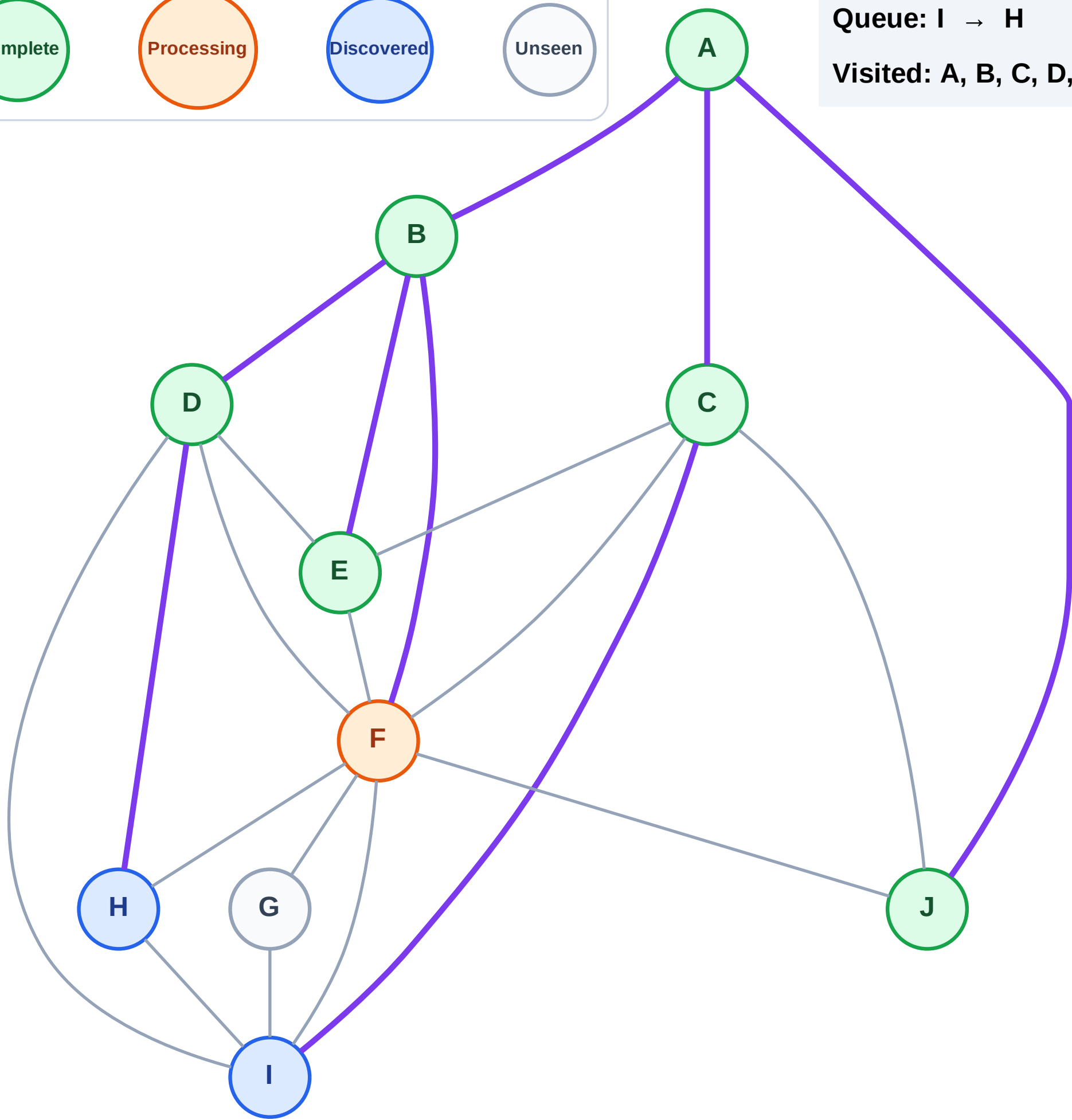
Step 14: Process node F

Legend



Queue: I → H

Visited: A, B, C, D, E, J



BFS Traversal

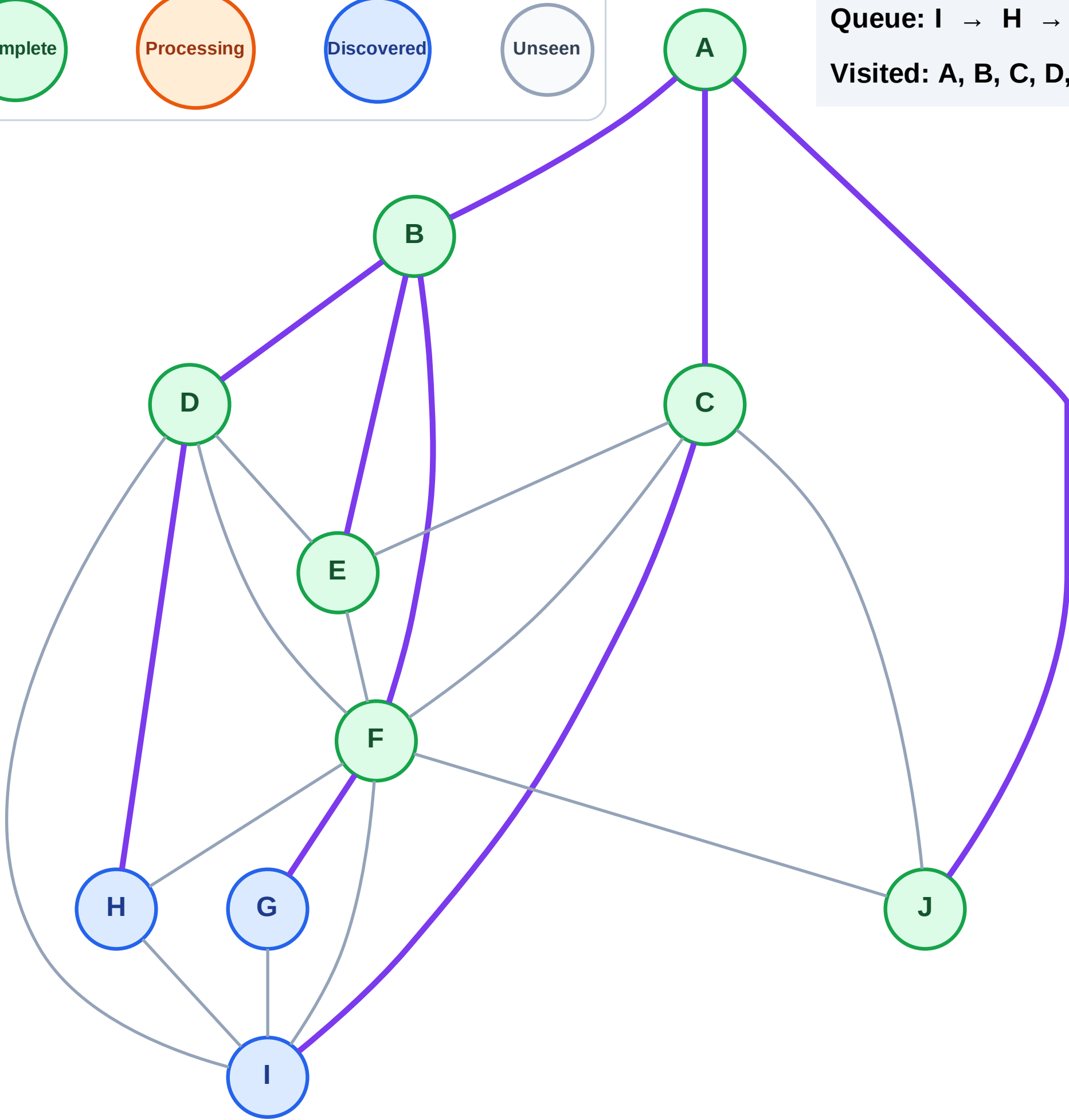
Step 15: Discover G from F

Legend



Queue: I → H → G

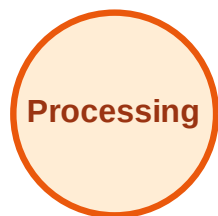
Visited: A, B, C, D, E, F, J



BFS Traversal

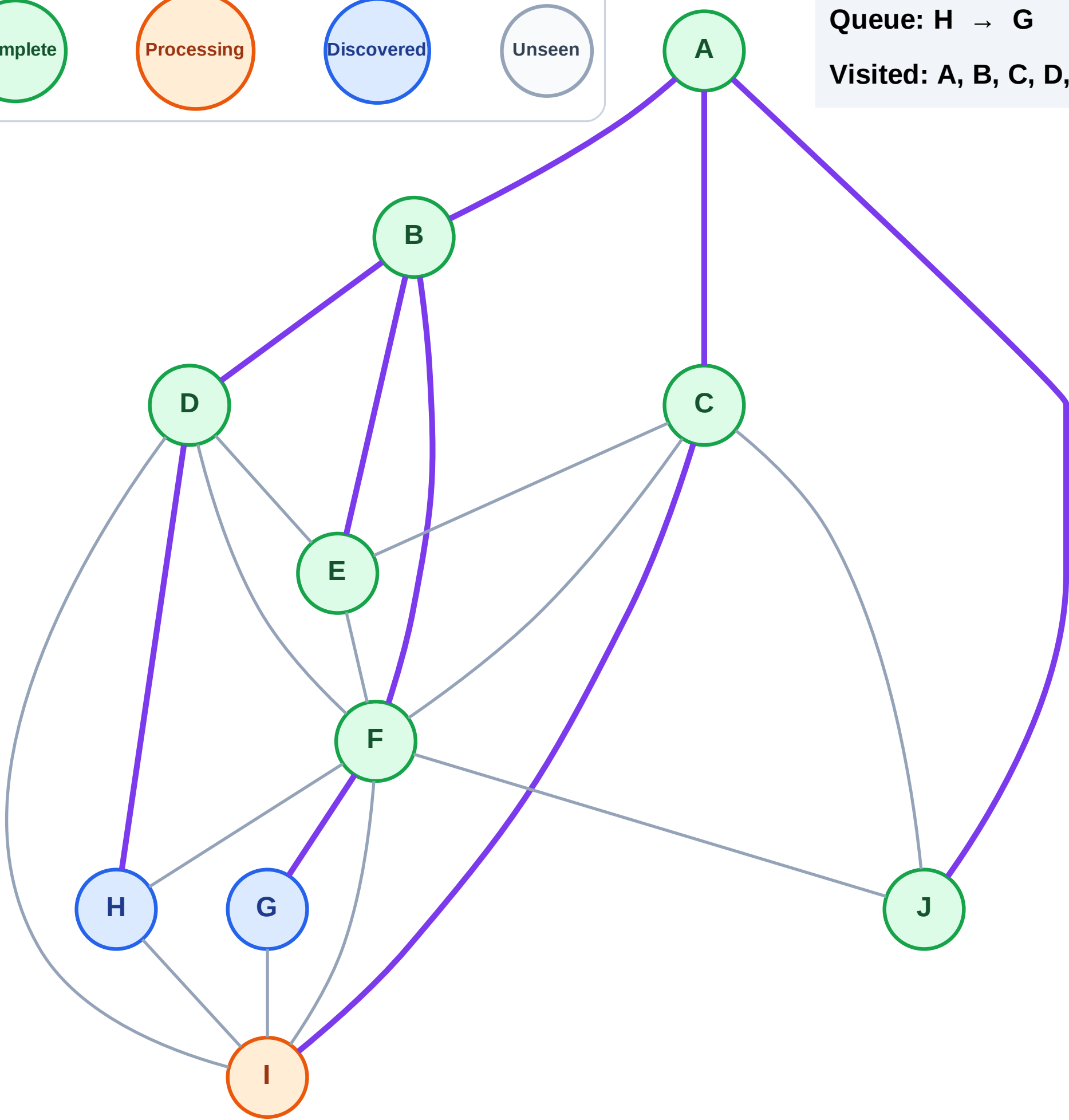
Step 16: Process node I

Legend



Queue: H → G

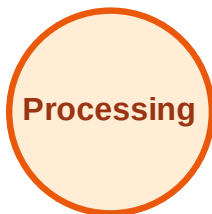
Visited: A, B, C, D, E, F, J



BFS Traversal

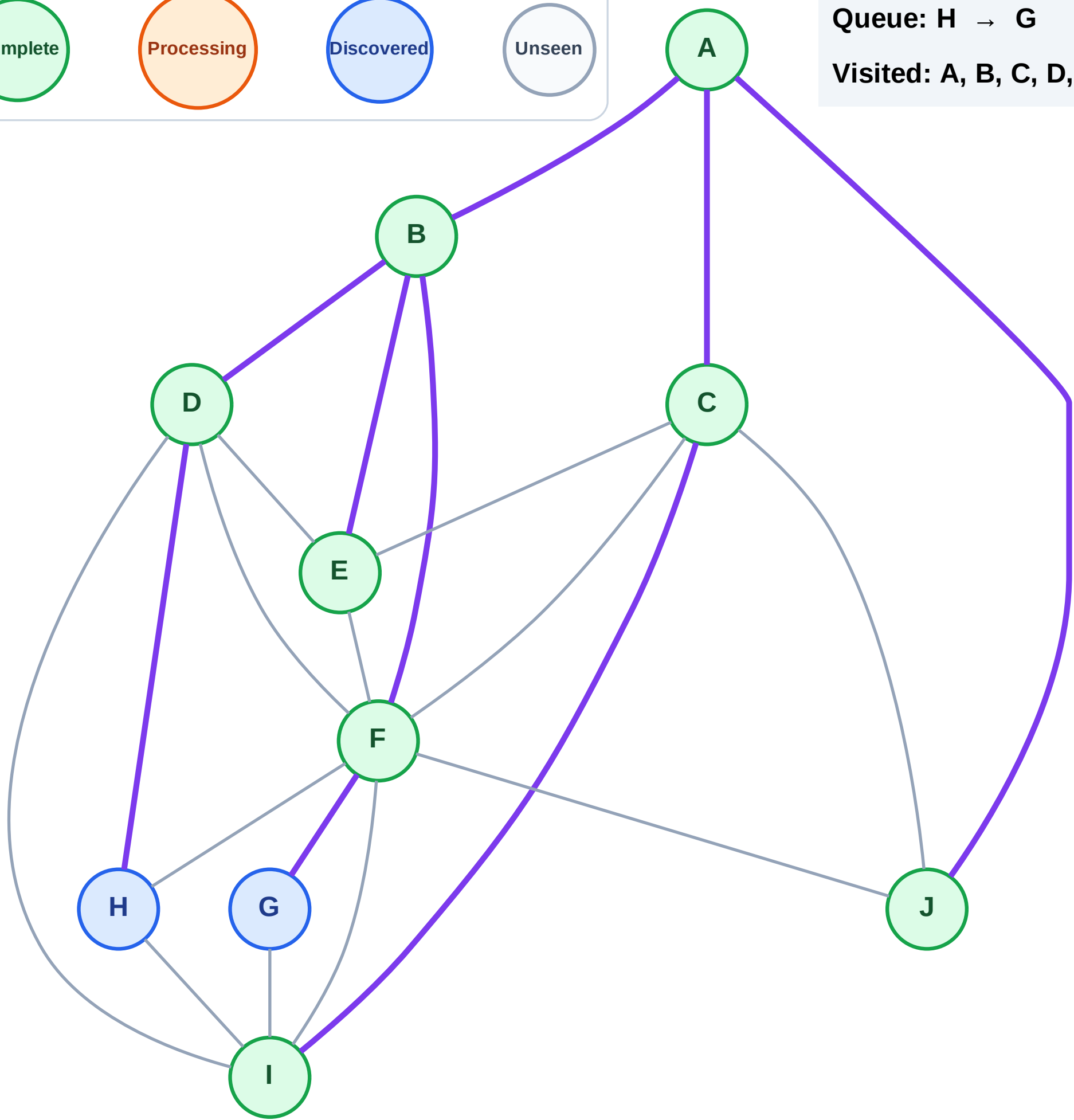
Step 17: No new neighbors from I

Legend



Queue: H → G

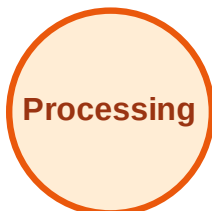
Visited: A, B, C, D, E, F, I, J



BFS Traversal

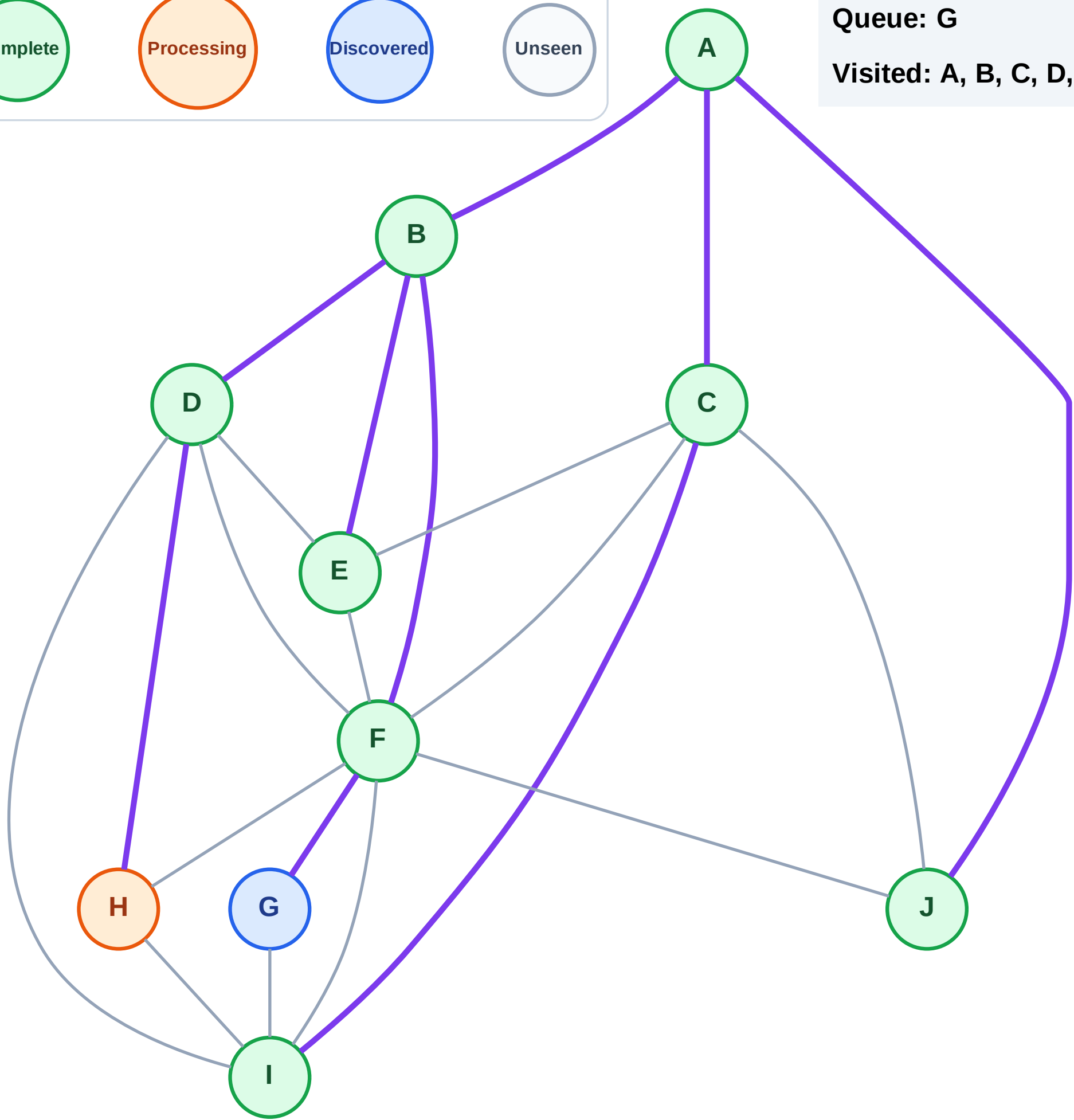
Step 18: Process node H

Legend



Queue: G

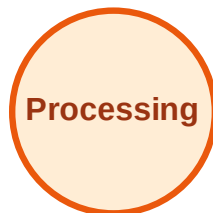
Visited: A, B, C, D, E, F, I, J



BFS Traversal

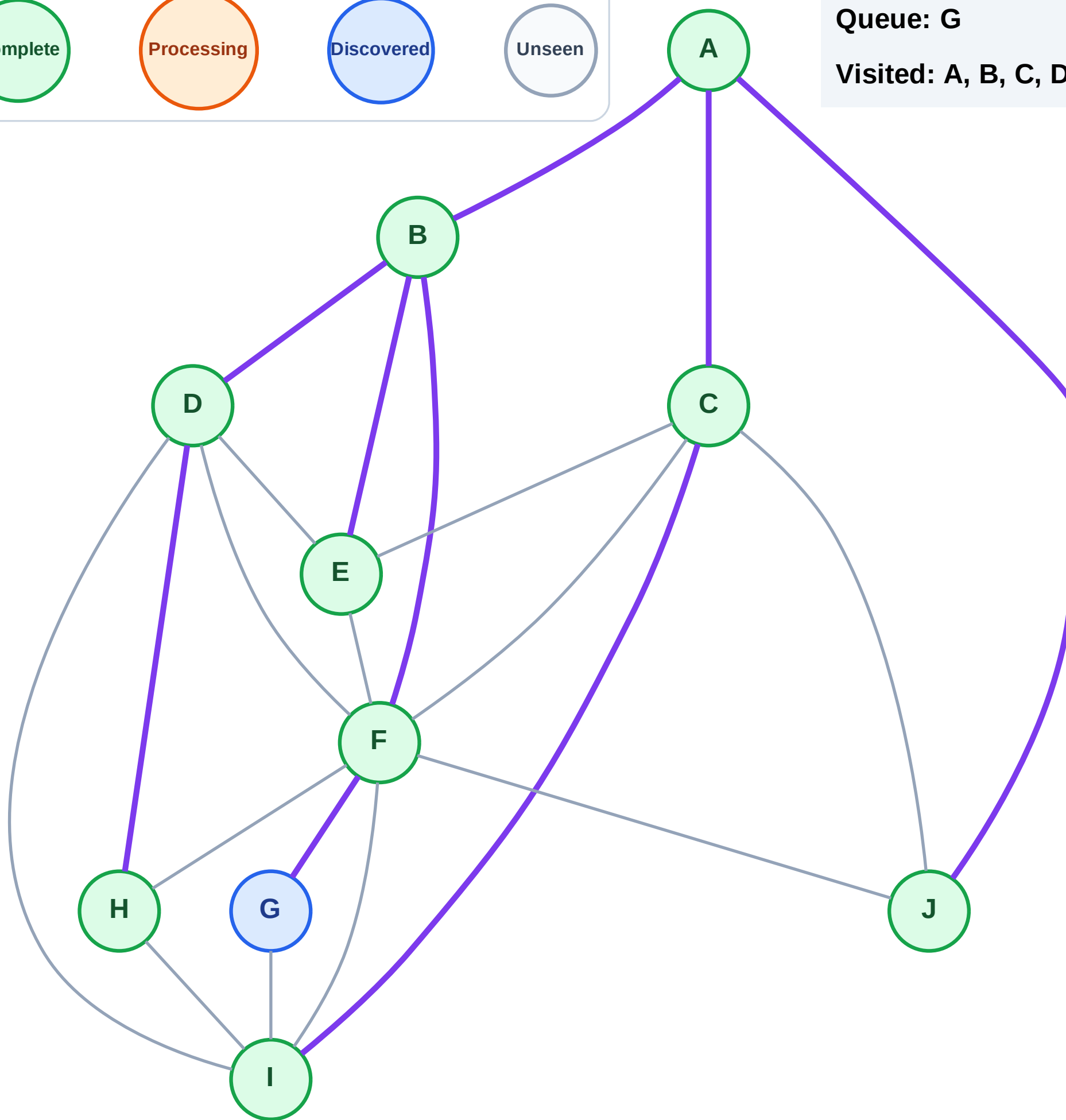
Step 19: No new neighbors from H

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

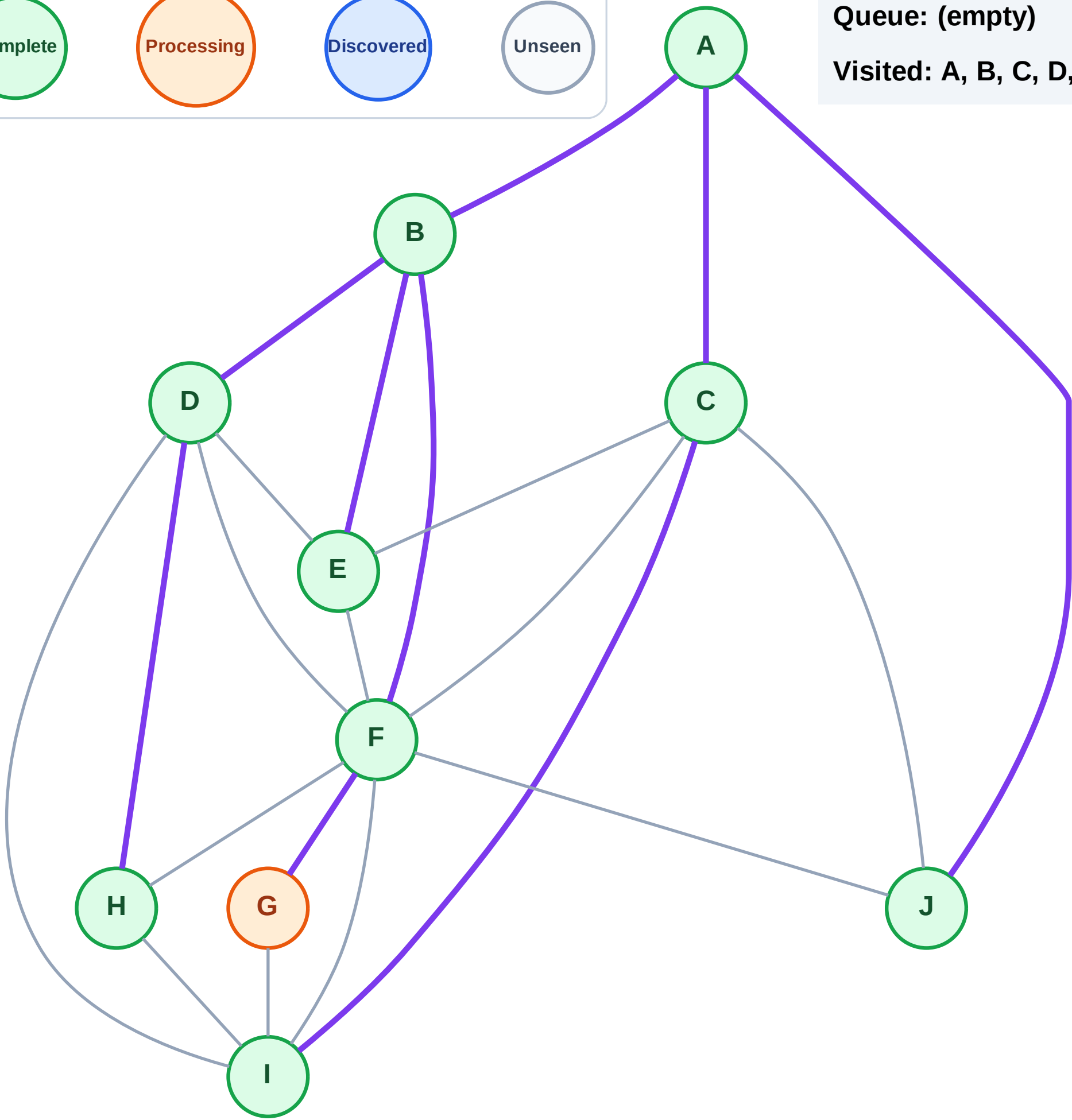
Step 20: Process node G

Legend



Queue: (empty)

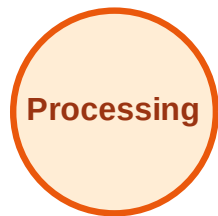
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

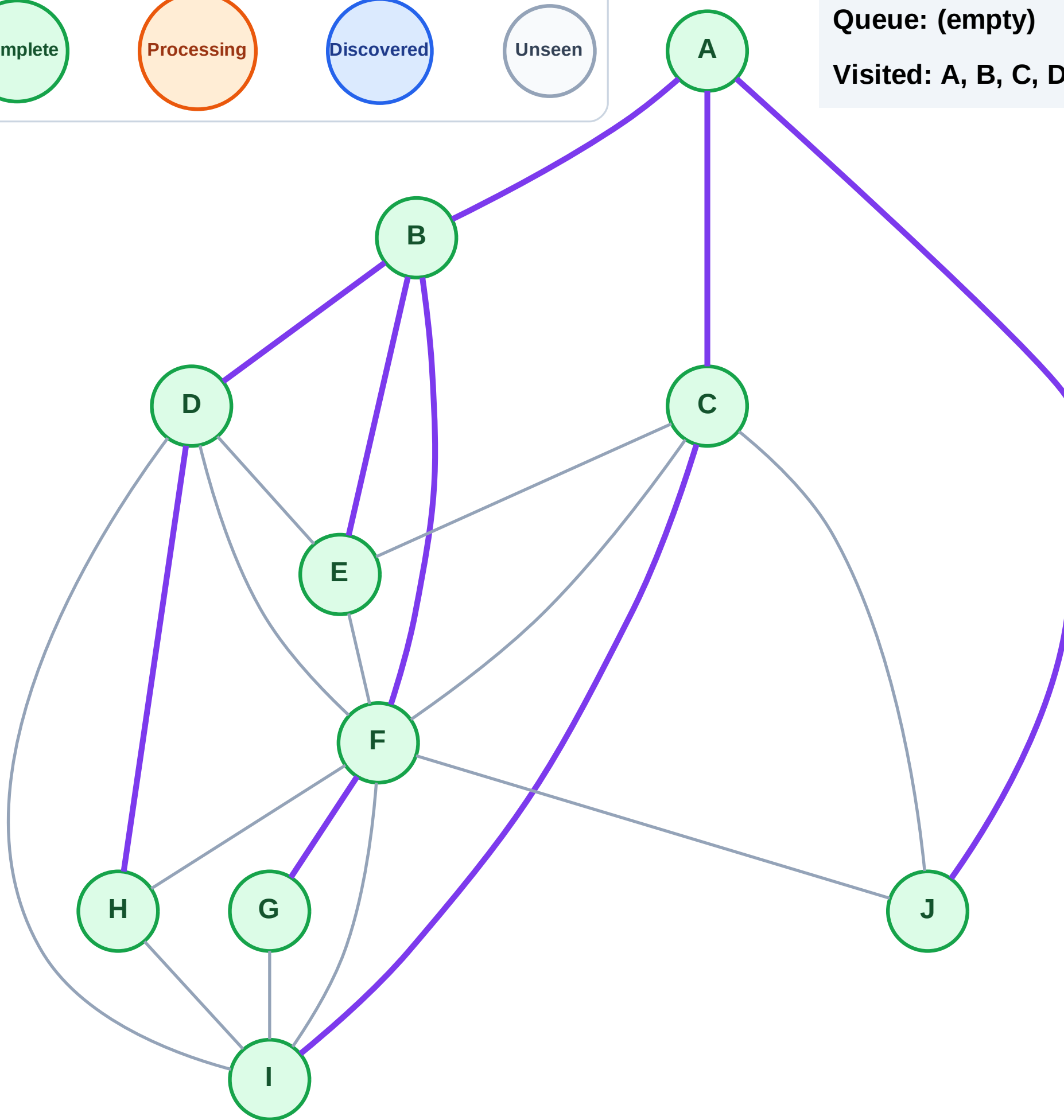
Step 21: No new neighbors from G

Legend



Queue: (empty)

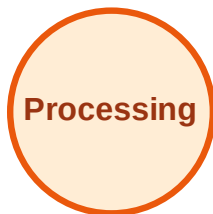
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

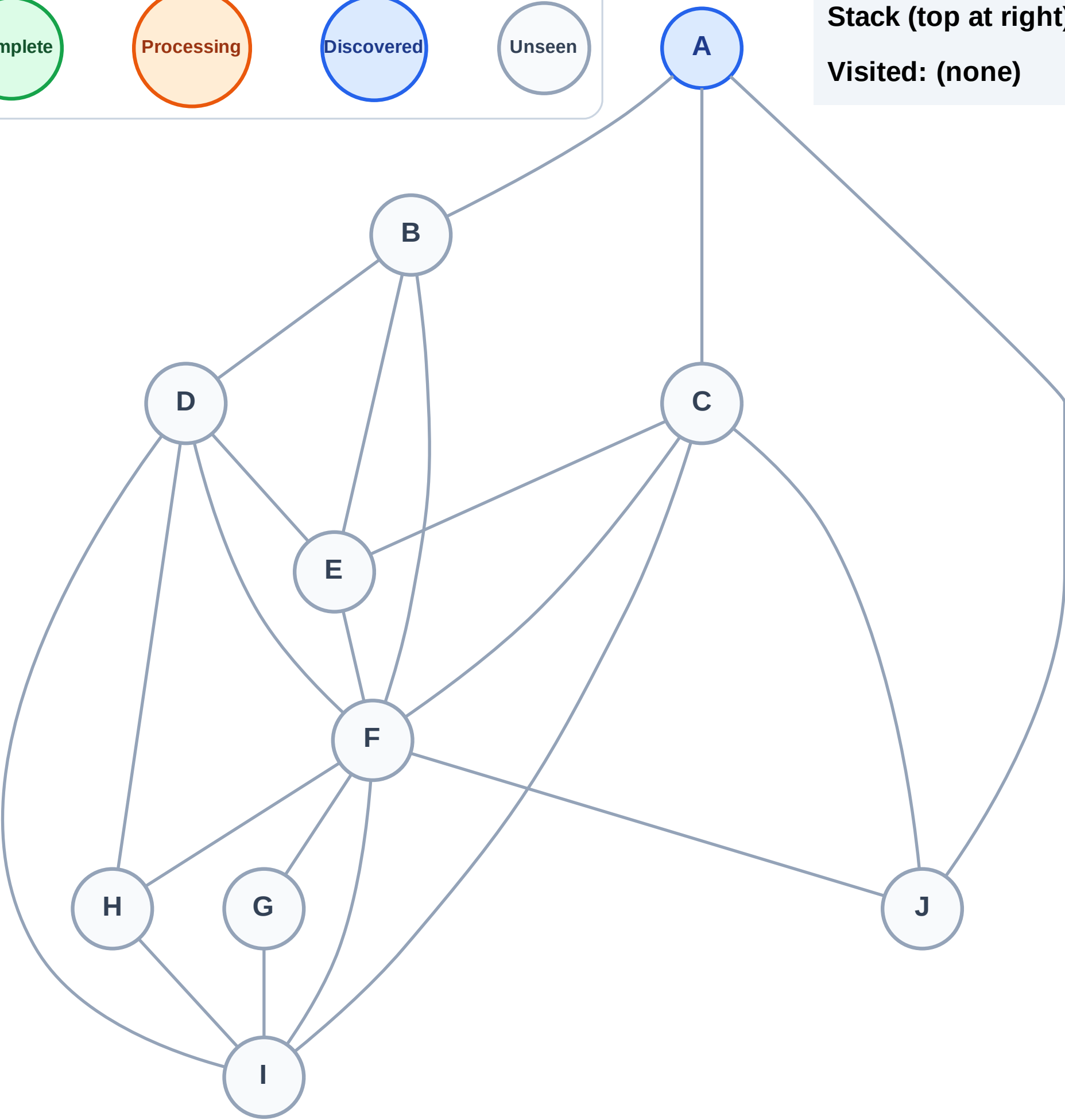
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

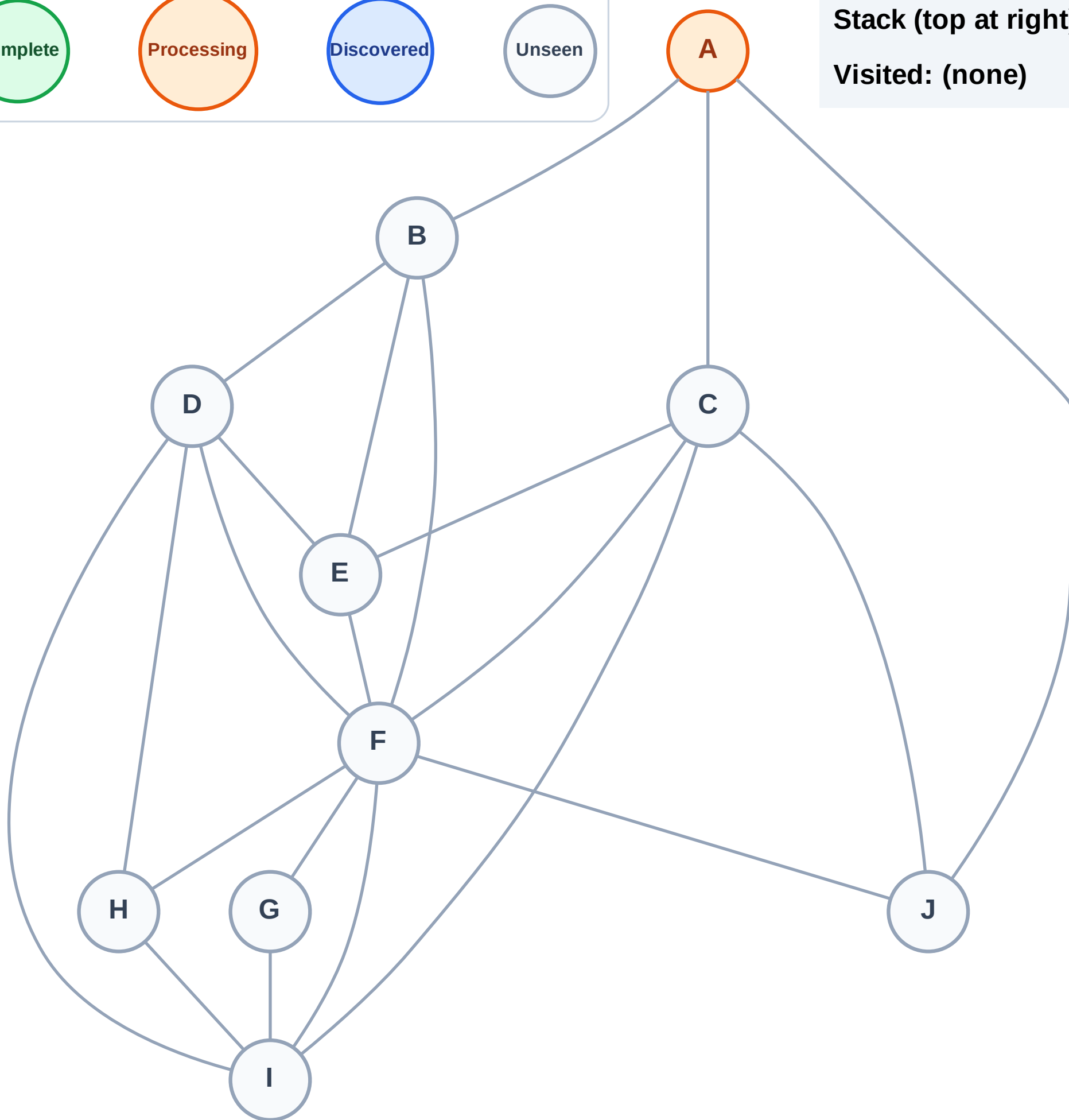
Processing

Discovered

Unseen

Stack (top at right): (empty)

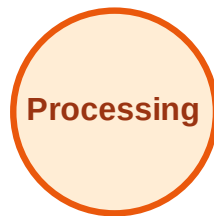
Visited: (none)



DFS Traversal

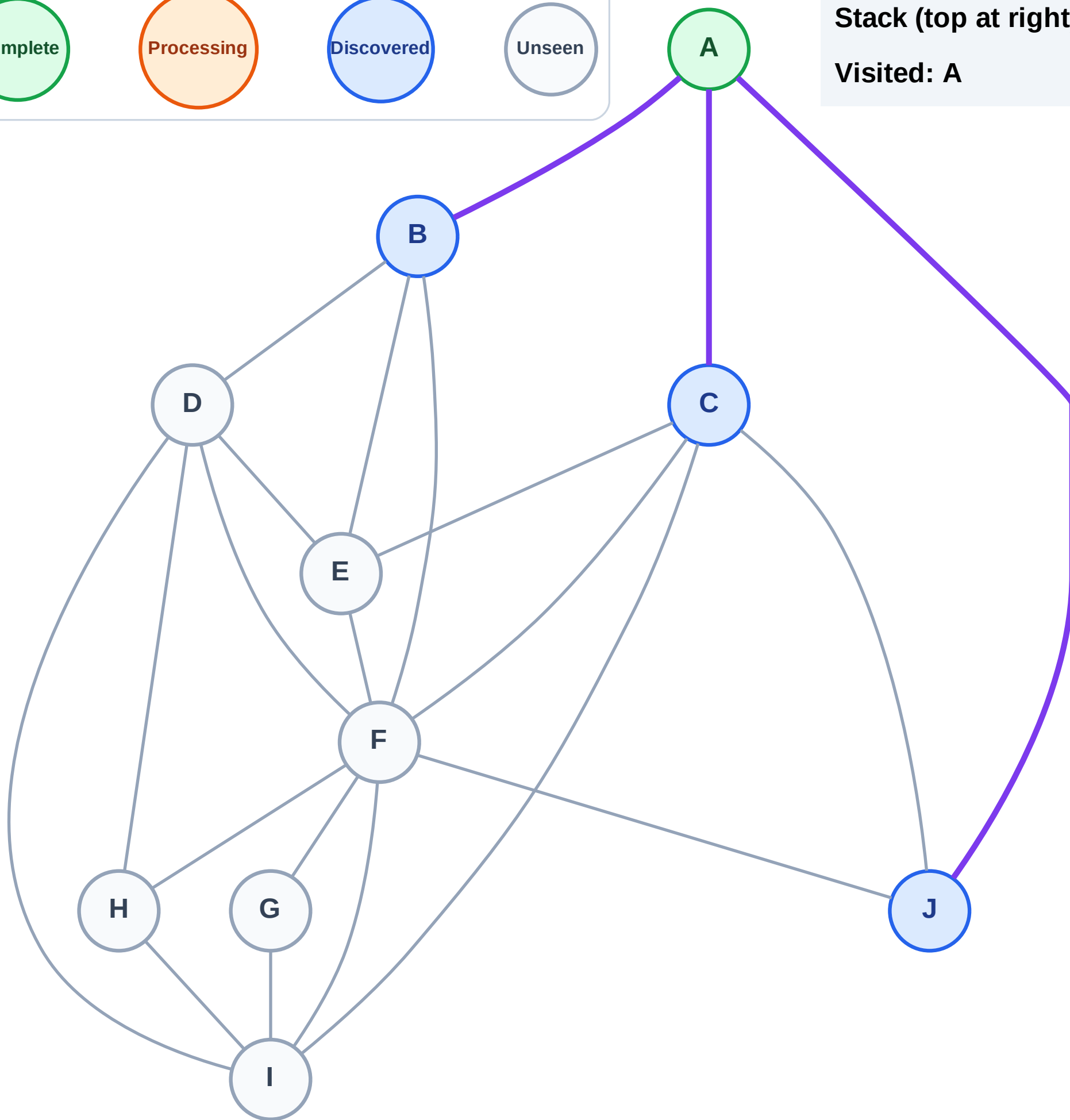
Step 3: Discover J, C, B from A

Legend



Stack (top at right): J | C | B

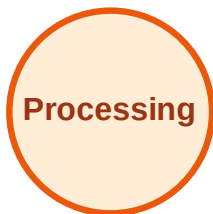
Visited: A



DFS Traversal

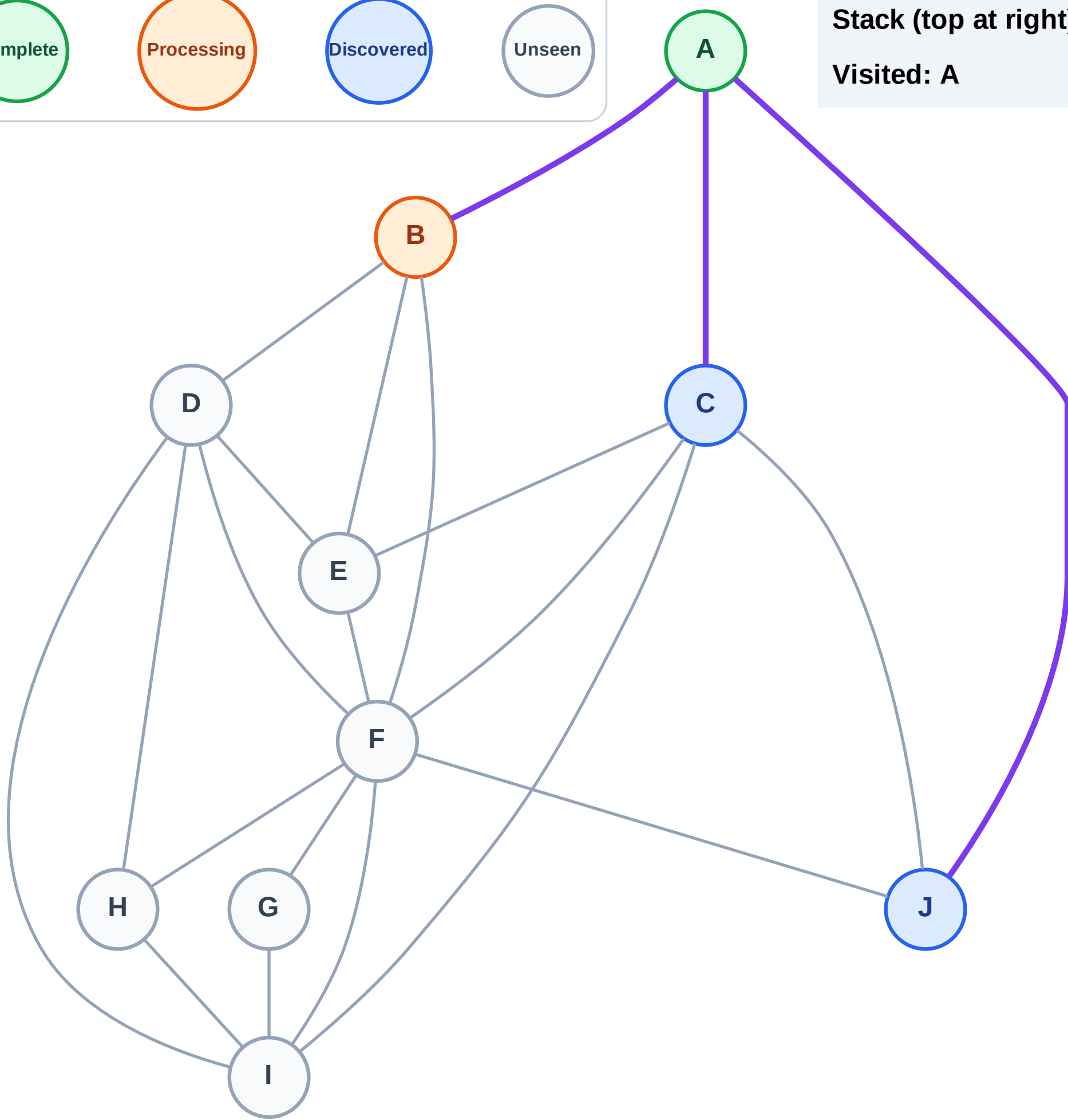
Step 4: Process node B

Legend



Stack (top at right): J | C

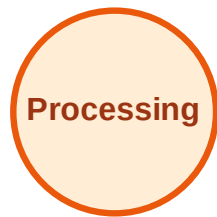
Visited: A



DFS Traversal

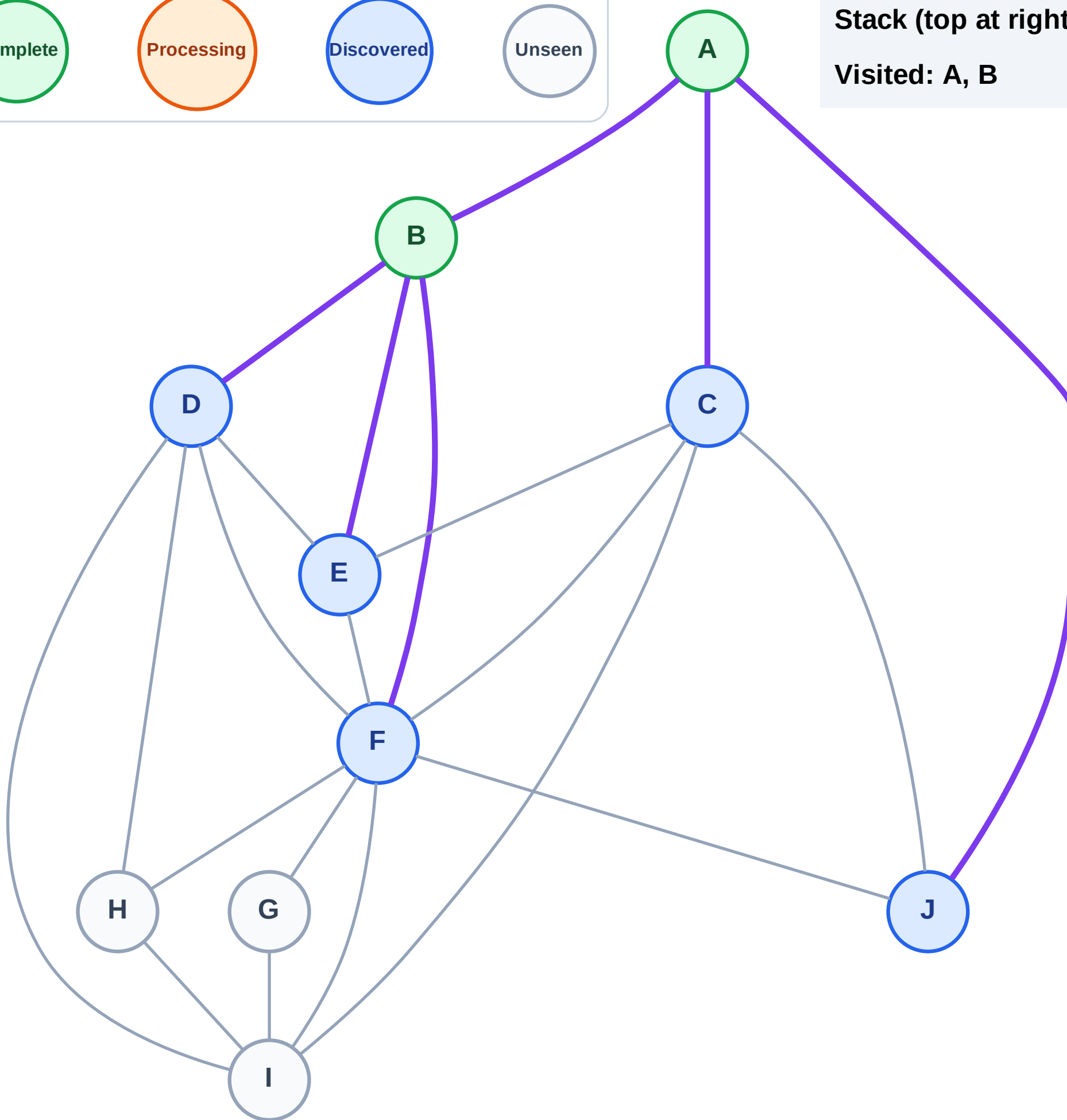
Step 5: Discover F, E, D from B

Legend



Stack (top at right): J | C | F | E | D

Visited: A, B



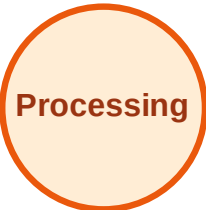
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



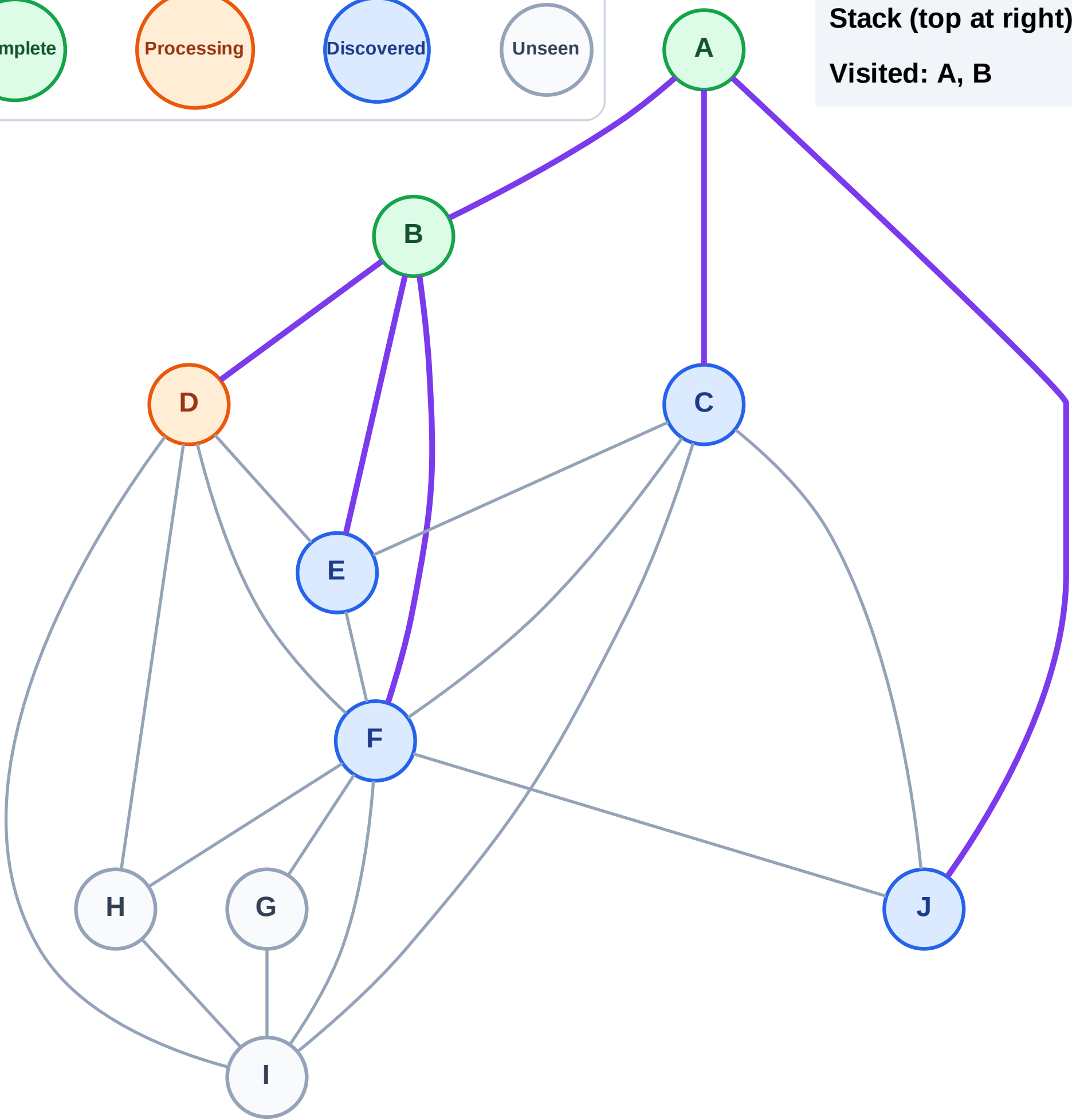
Discovered



Unseen

Stack (top at right): J | C | F | E

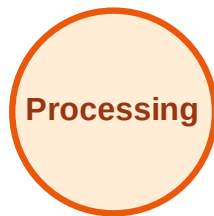
Visited: A, B



DFS Traversal

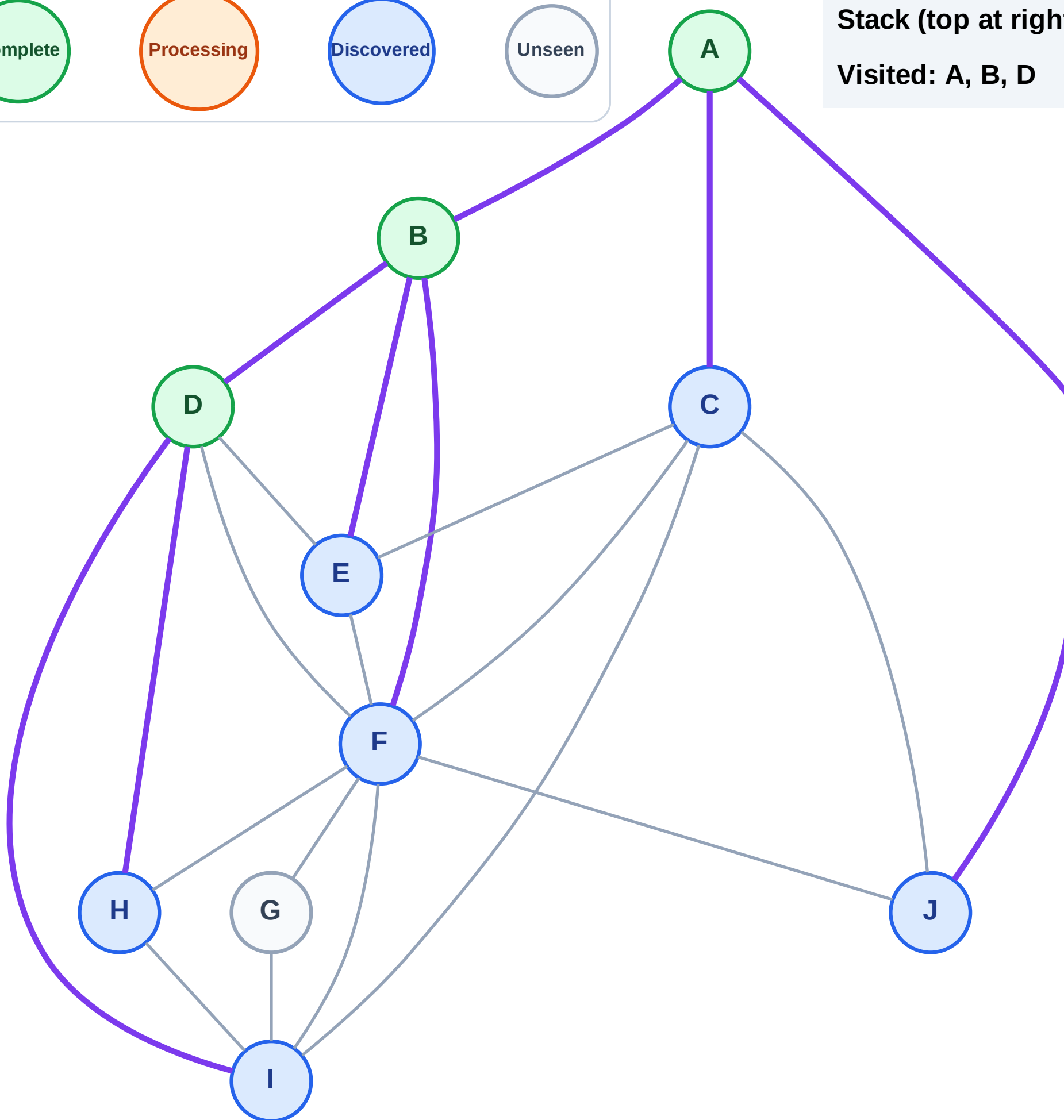
Step 7: Discover I, H from D

Legend



Stack (top at right): J | C | F | E | I | H

Visited: A, B, D



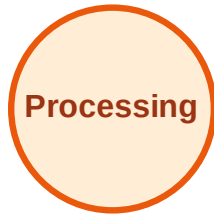
DFS Traversal

Step 8: Process node H

Legend



Complete



Processing



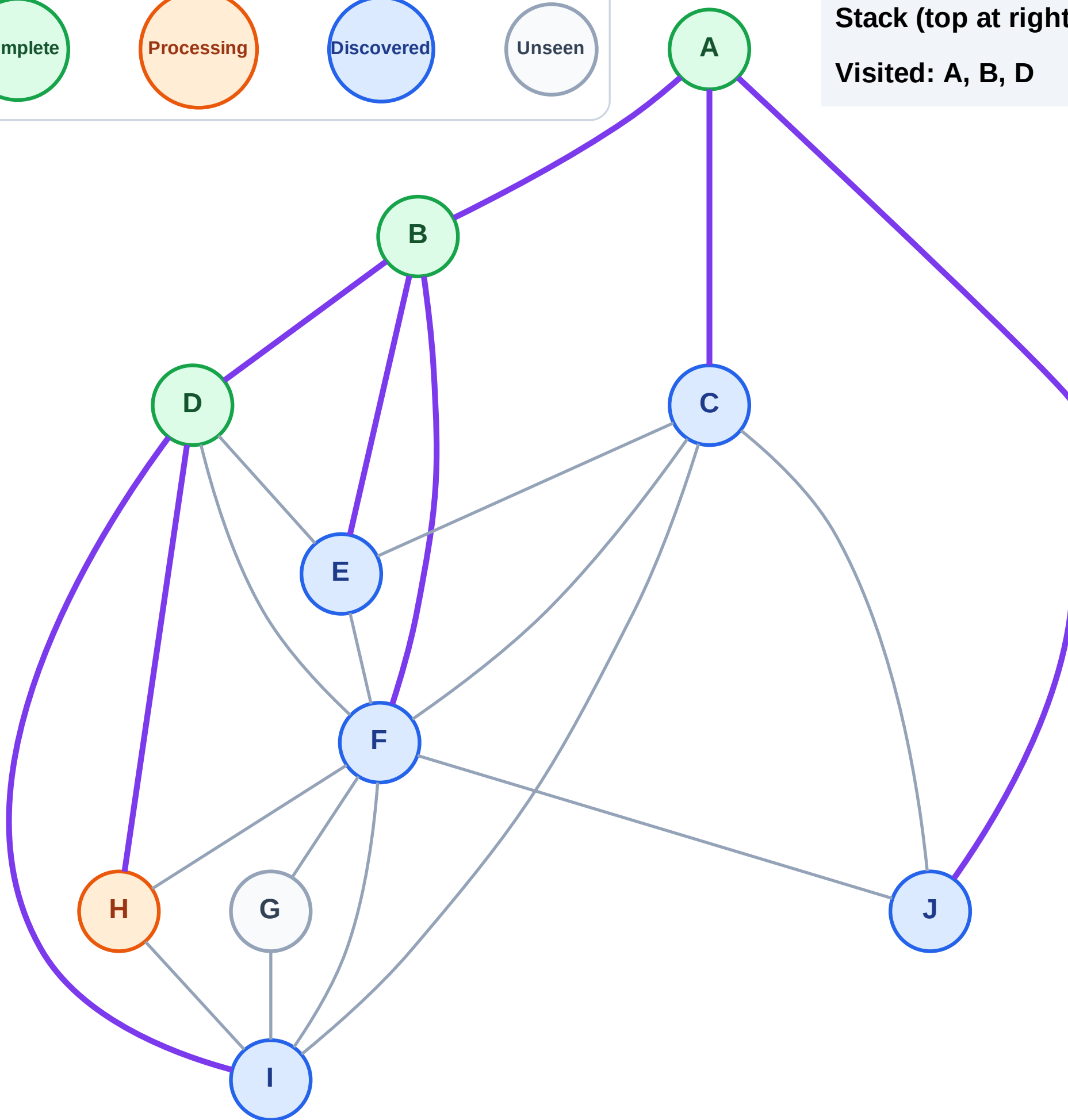
Discovered



Unseen

Stack (top at right): J | C | F | E | I

Visited: A, B, D



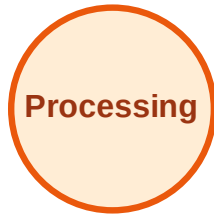
DFS Traversal

Step 9: No new neighbors from H

Legend



Complete



Processing



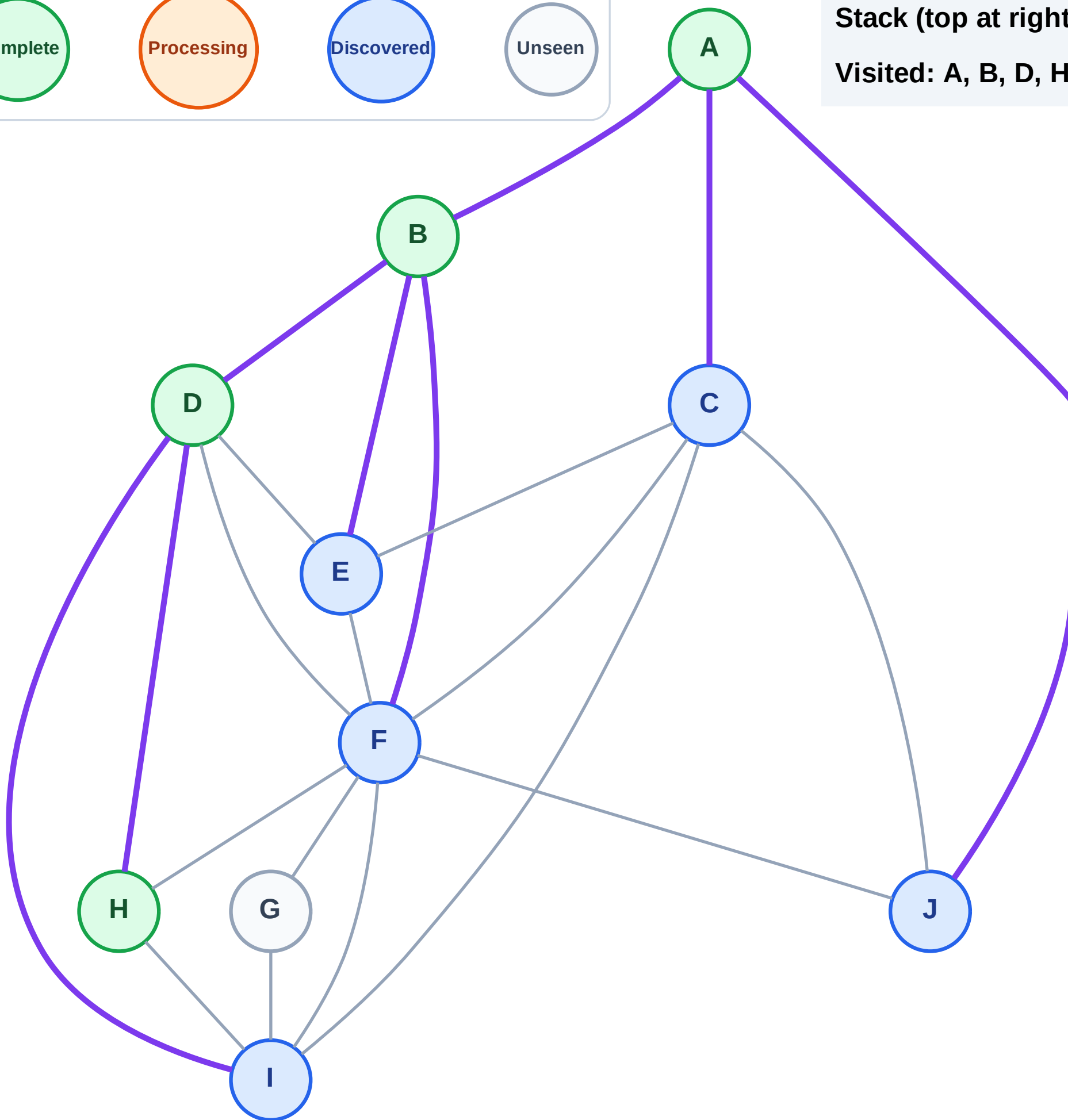
Discovered



Unseen

Stack (top at right): J | C | F | E | I

Visited: A, B, D, H



DFS Traversal

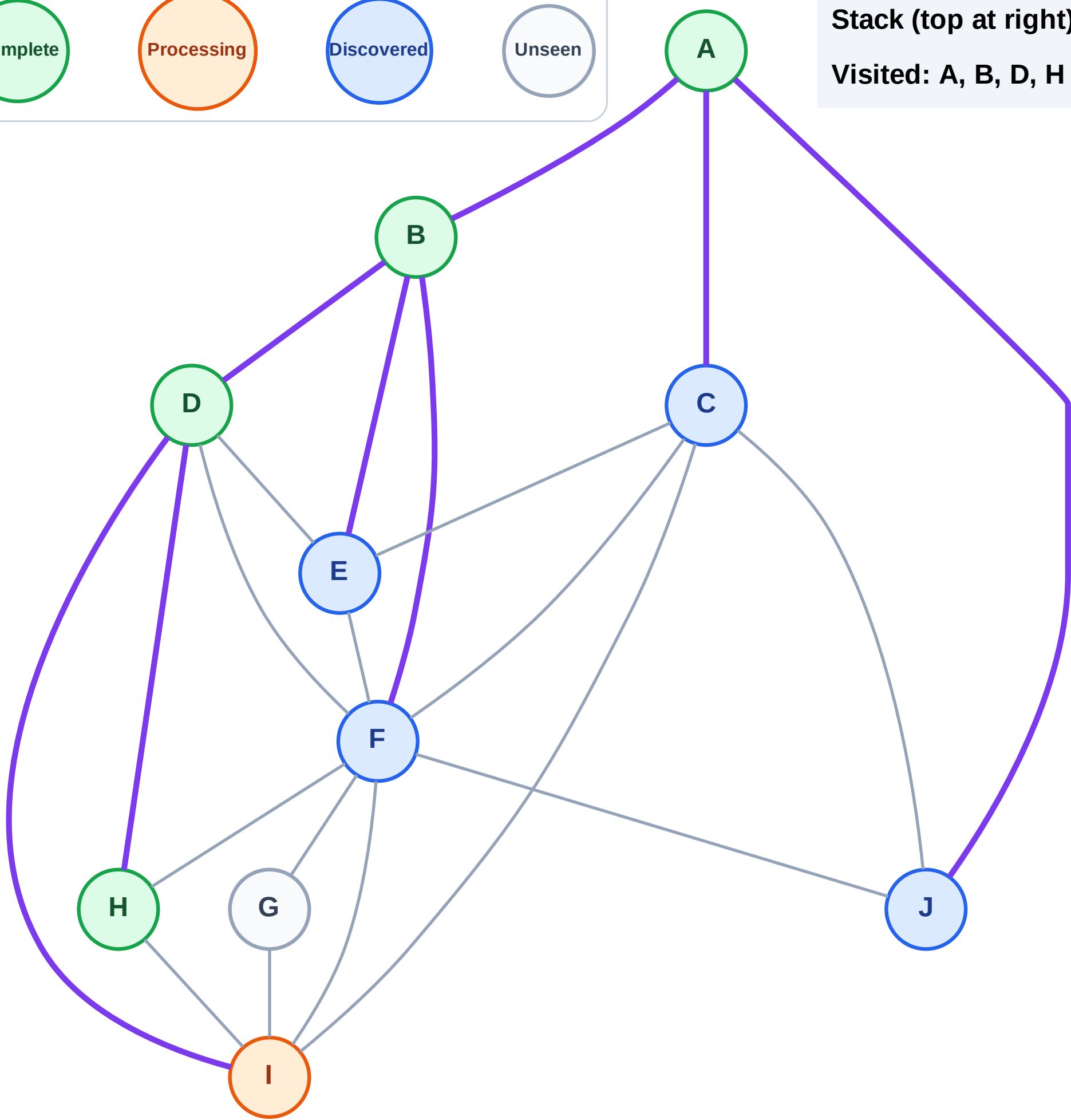
Step 10: Process node I

Legend



Stack (top at right): J | C | F | E

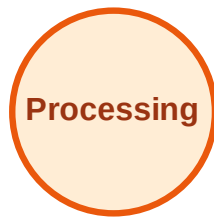
Visited: A, B, D, H



DFS Traversal

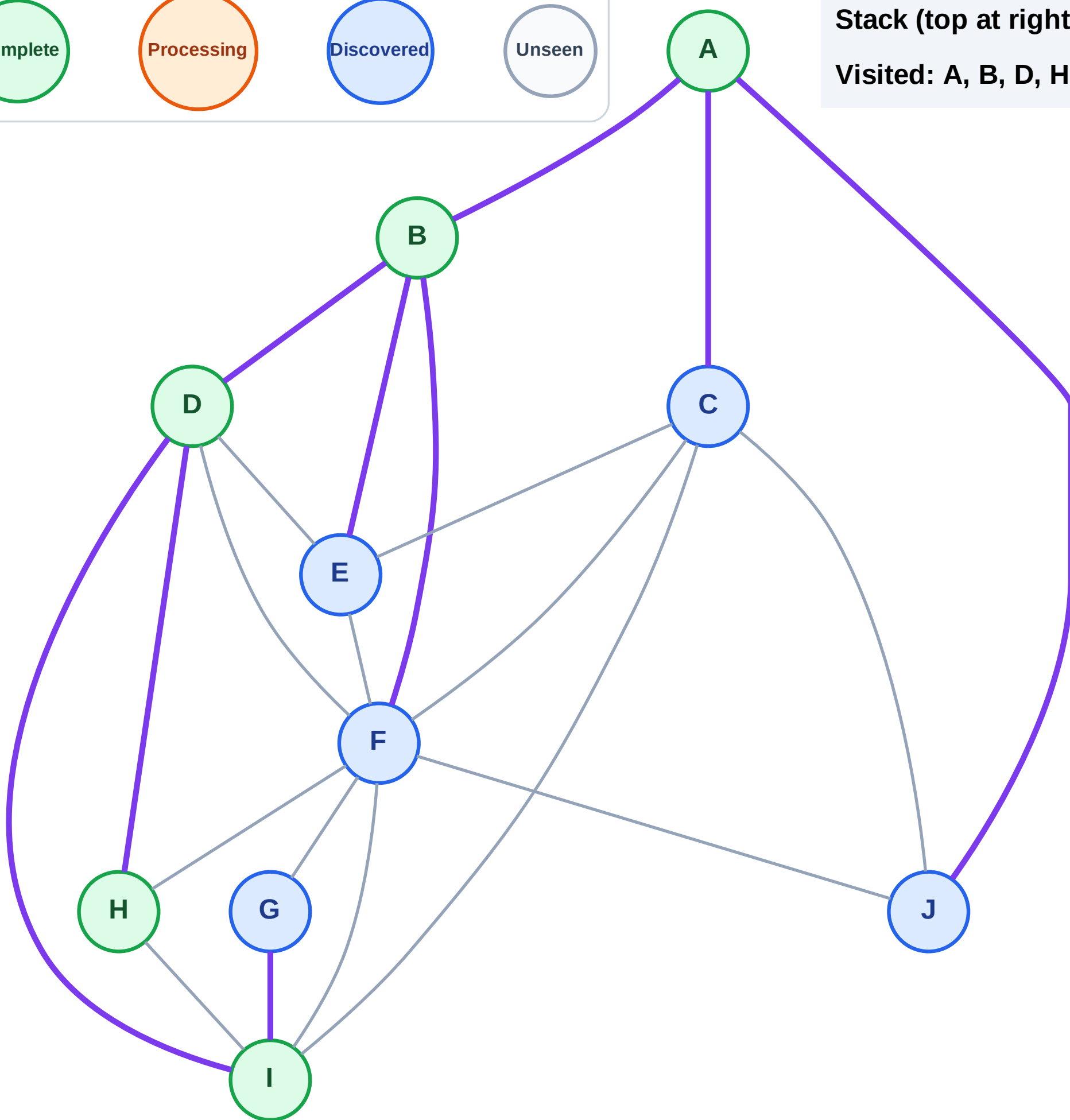
Step 11: Discover G from I

Legend



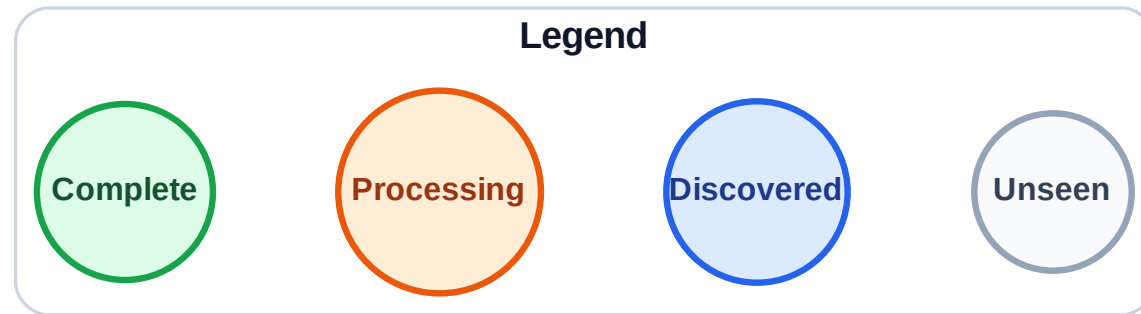
Stack (top at right): J | C | F | E | G

Visited: A, B, D, H, I



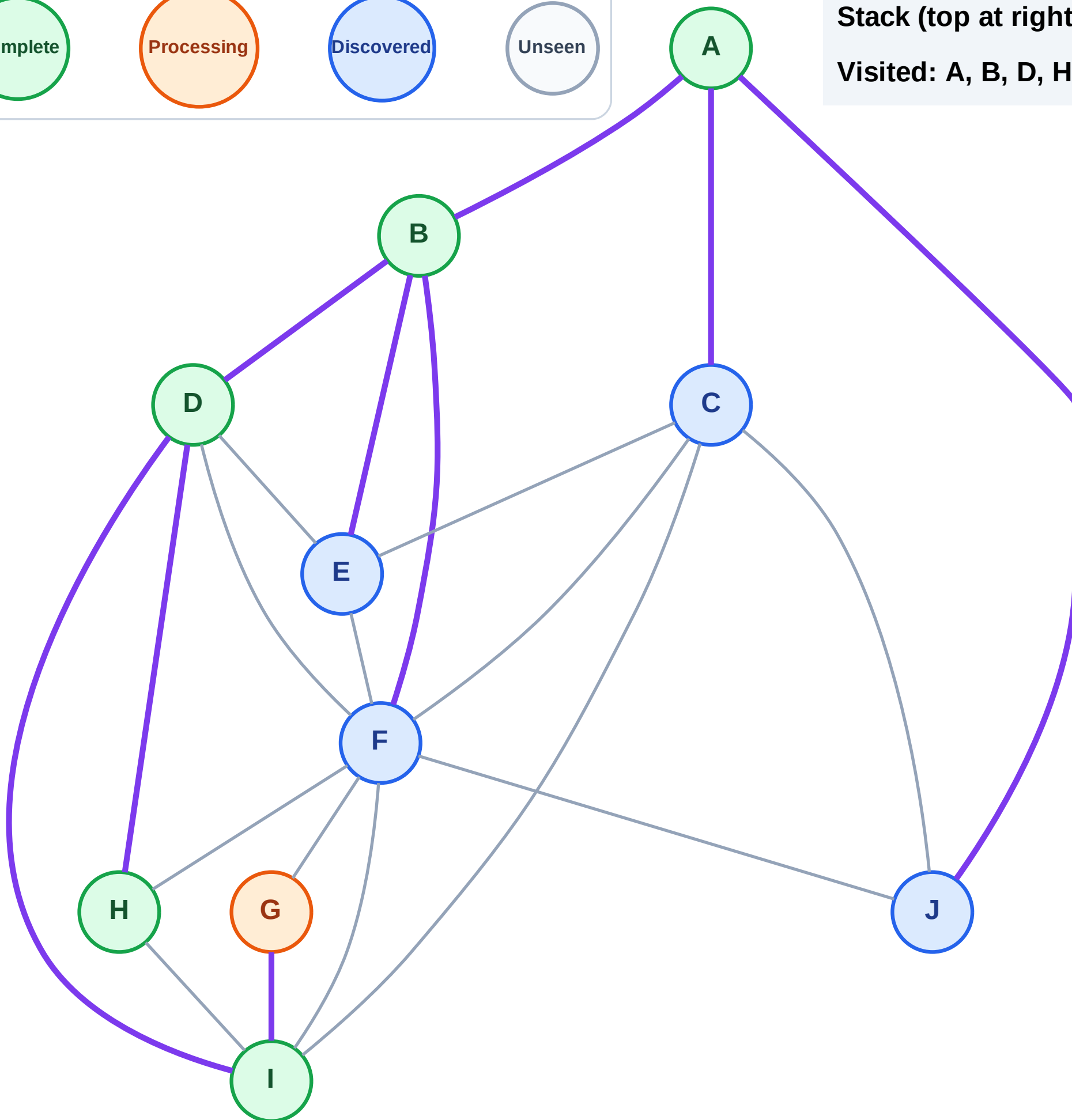
DFS Traversal

Step 12: Process node G



Stack (top at right): J | C | F | E

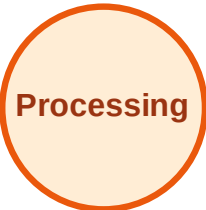
Visited: A, B, D, H, I



DFS Traversal

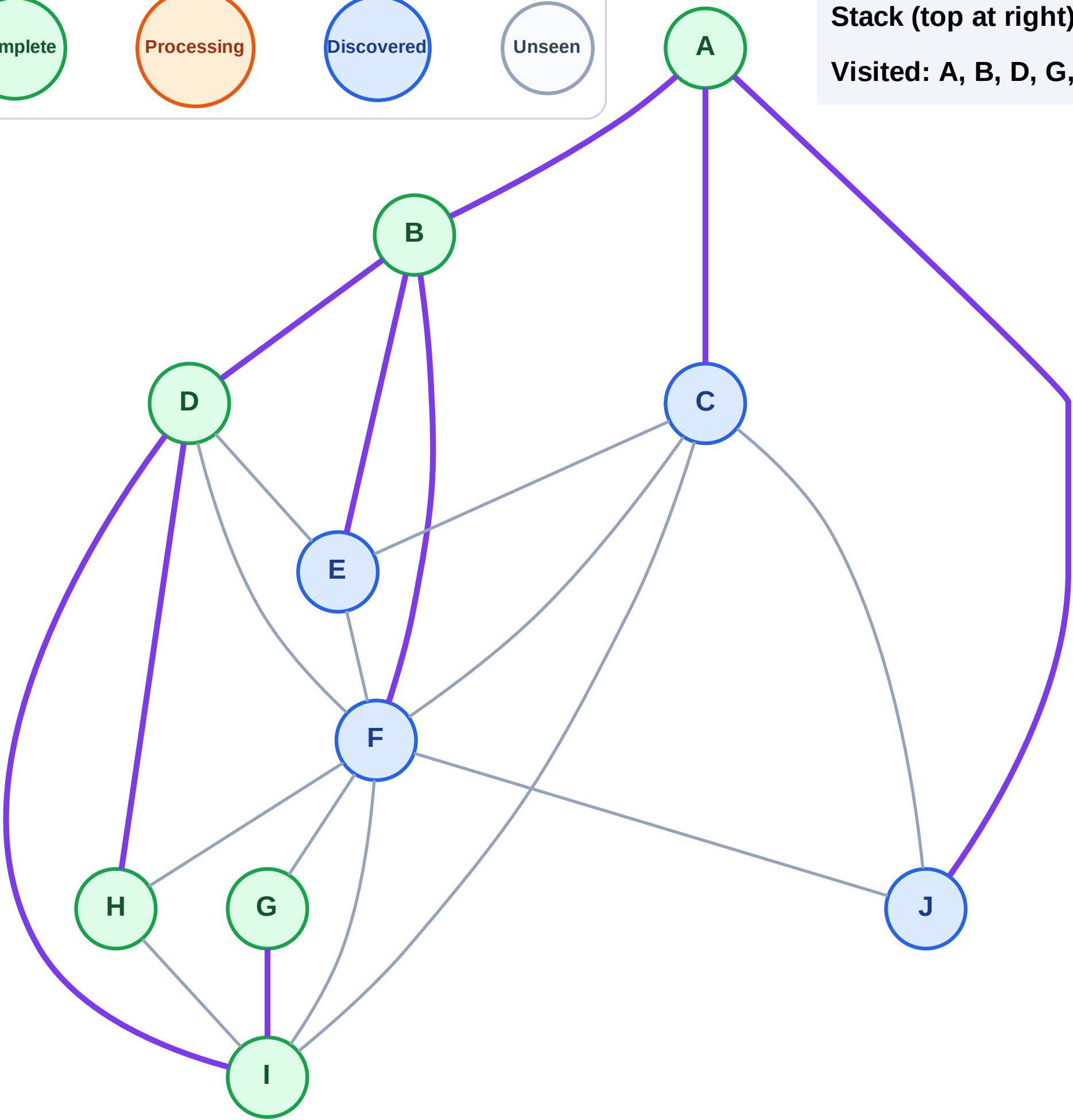
Step 13: No new neighbors from G

Legend



Stack (top at right): J | C | F | E

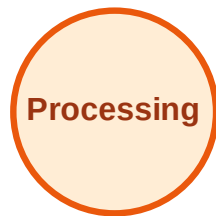
Visited: A, B, D, G, H, I



DFS Traversal

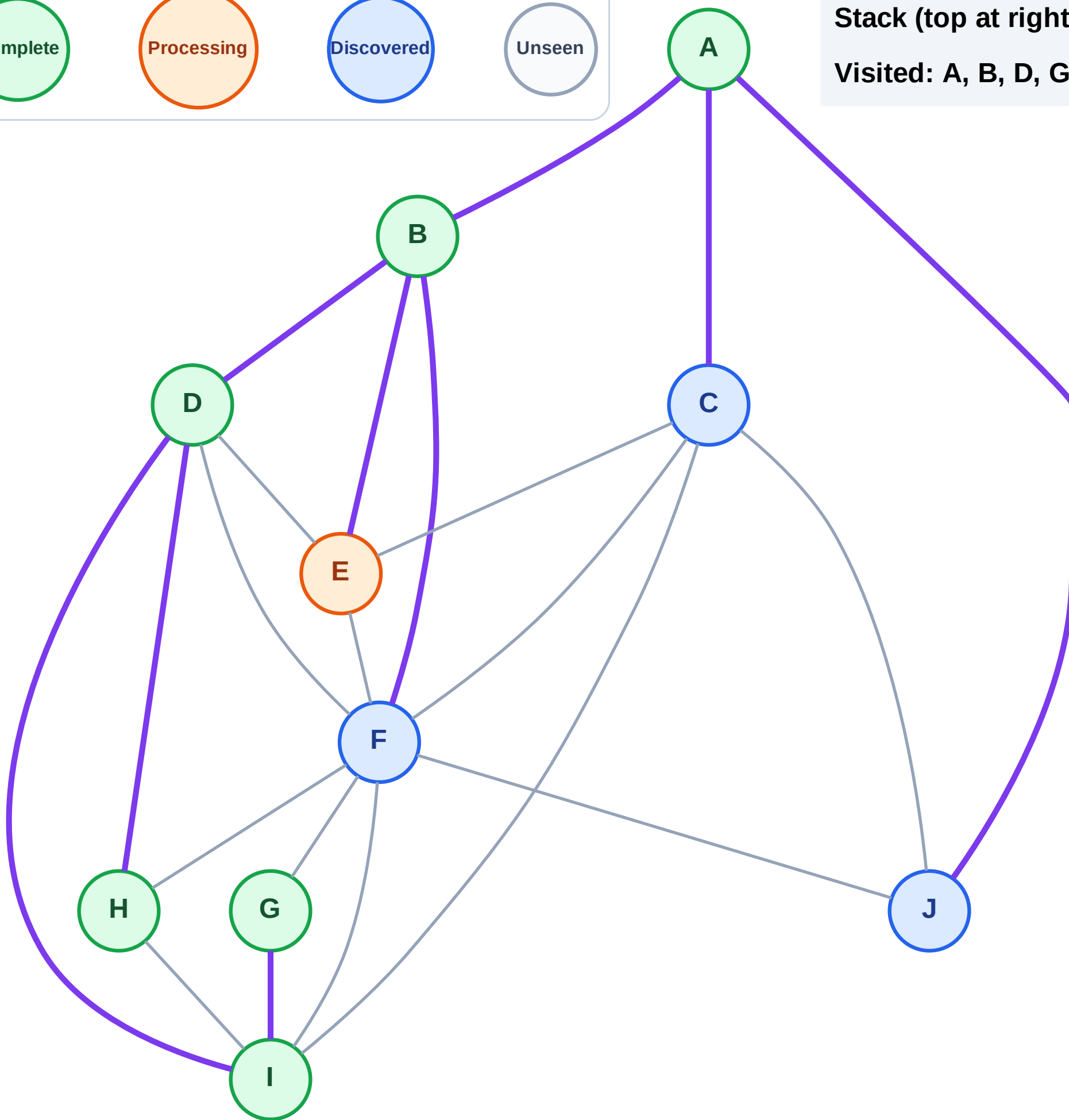
Step 14: Process node E

Legend



Stack (top at right): J | C | F

Visited: A, B, D, G, H, I

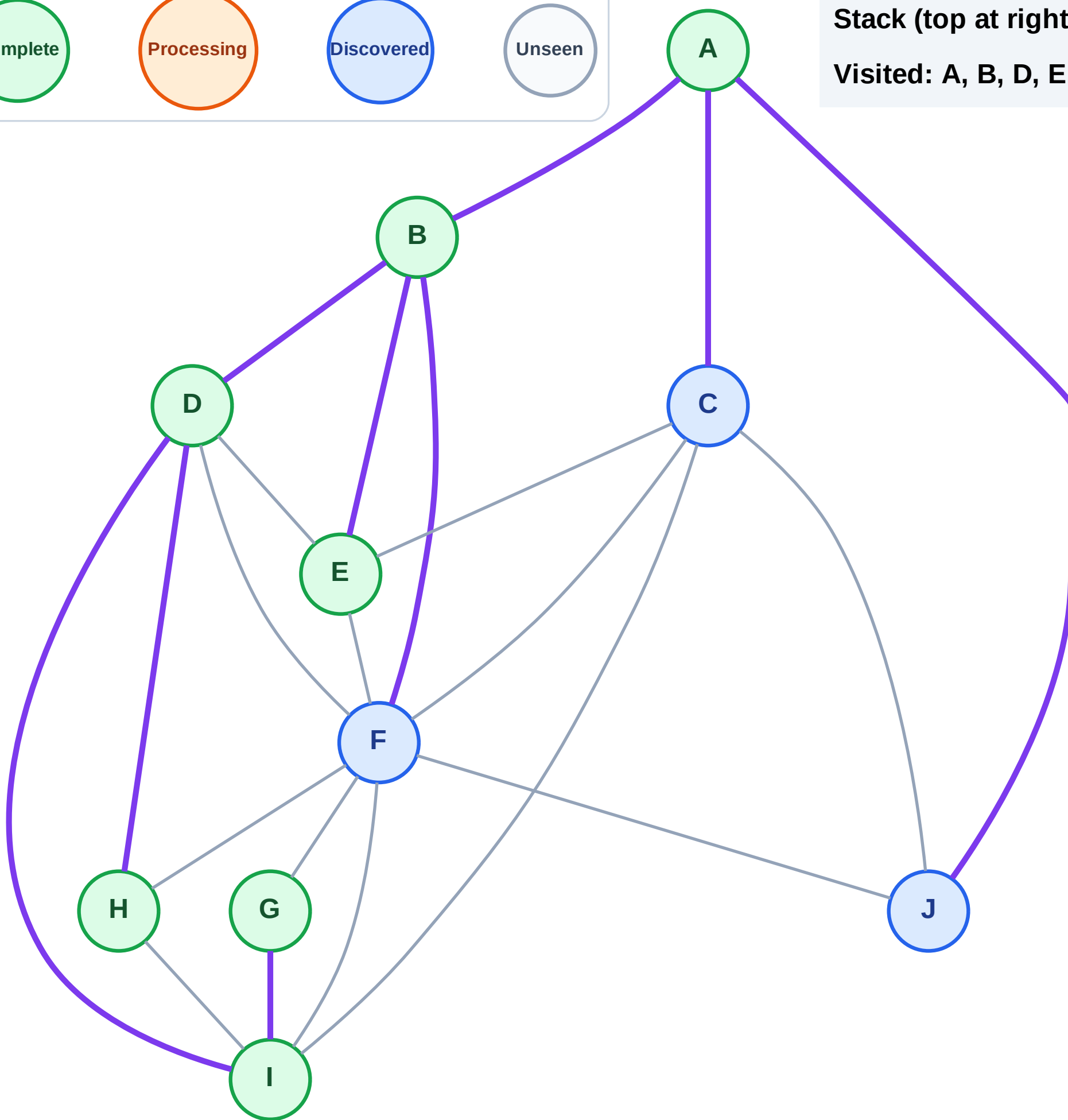


Step 15: No new neighbors from E

Legend

Unseen

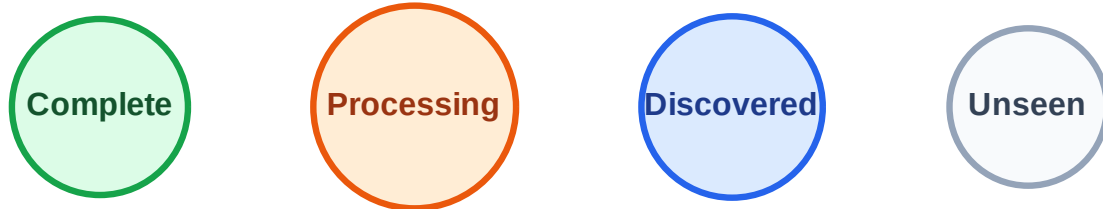
Visited: A, B, D, E, G, H, I



DFS Traversal

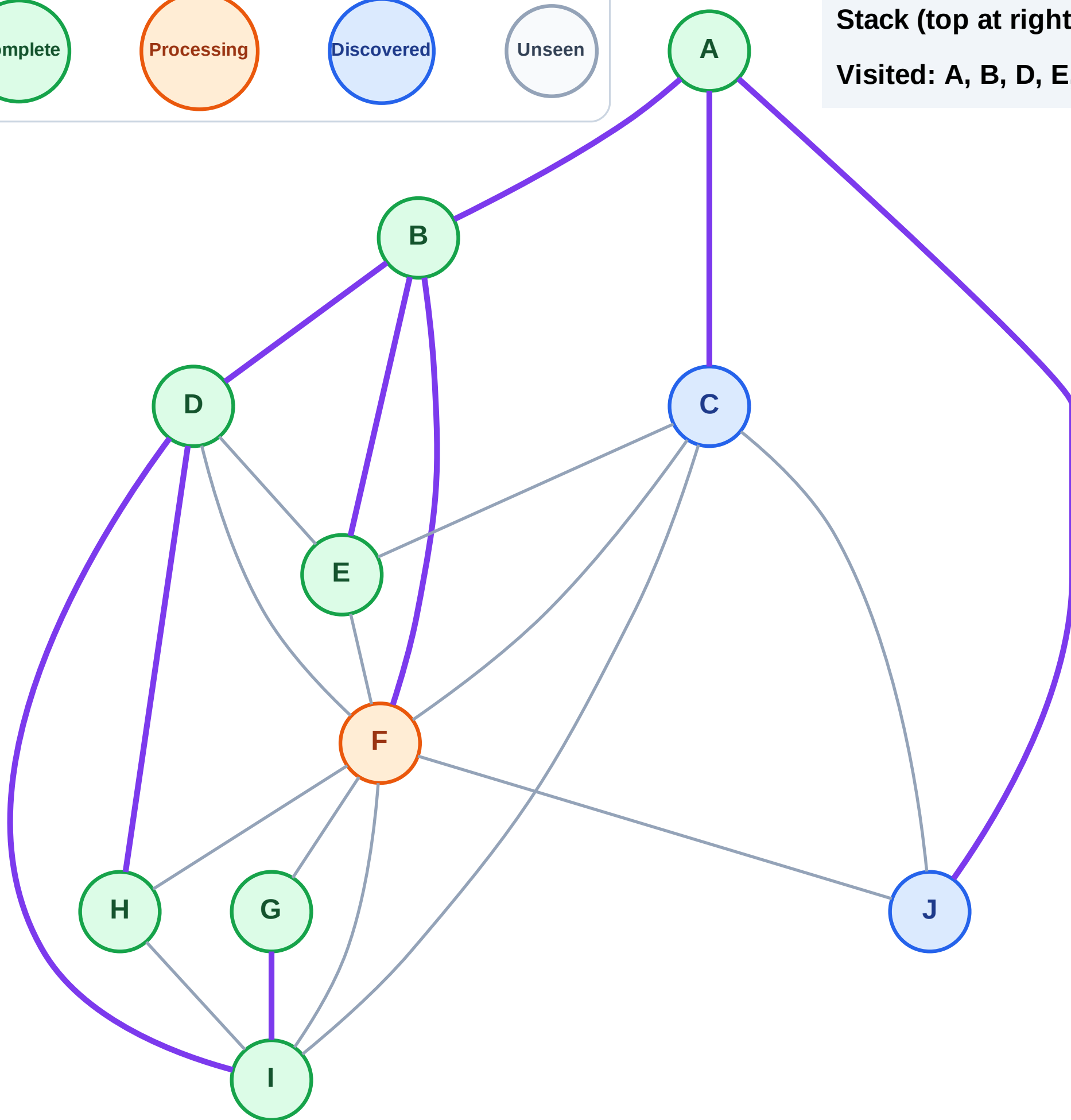
Step 16: Process node F

Legend



Stack (top at right): J | C

Visited: A, B, D, E, G, H, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

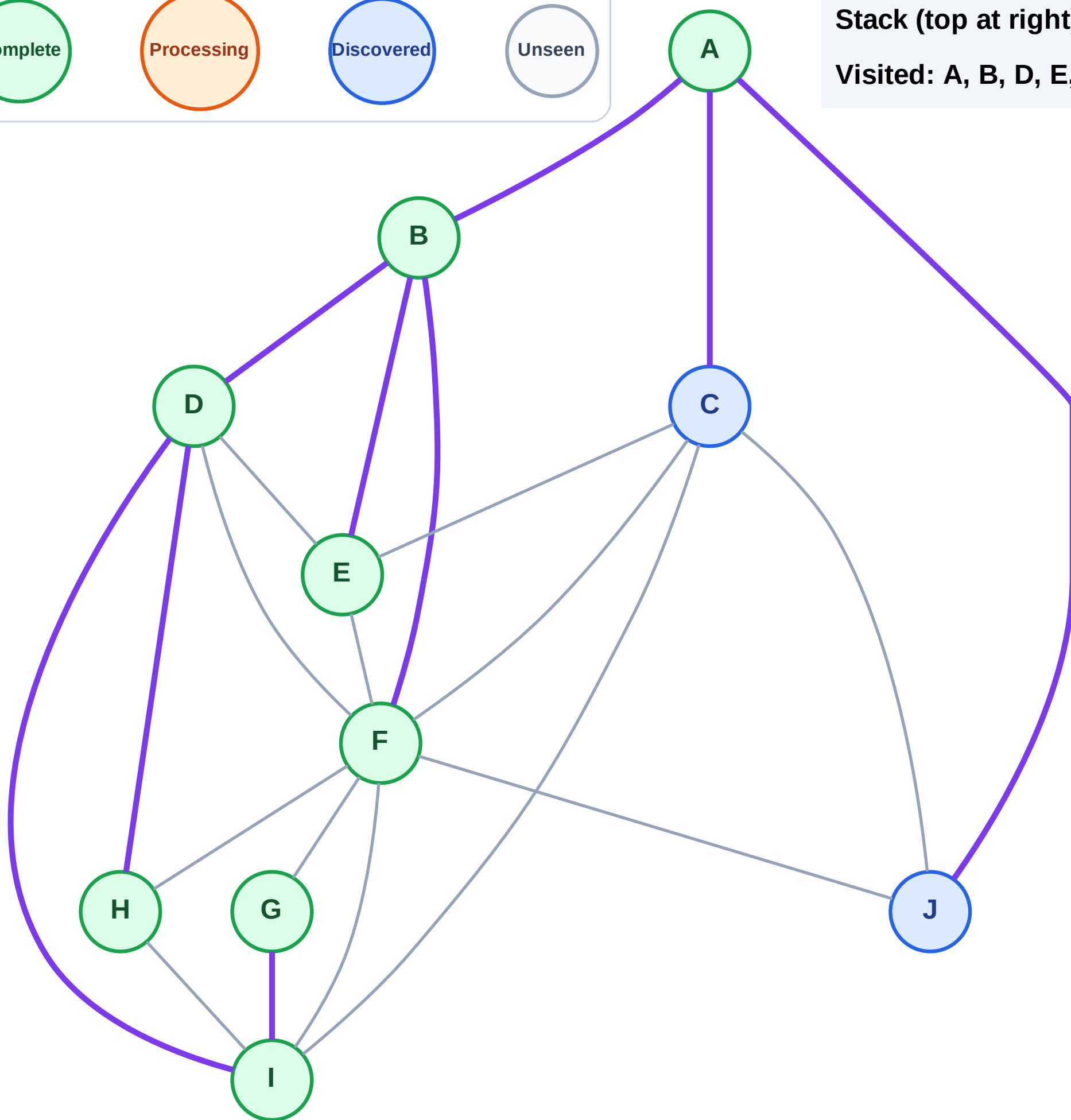
Processing

Discovered

Unseen

Stack (top at right): J | C

Visited: A, B, D, E, F, G, H, I



DFS Traversal

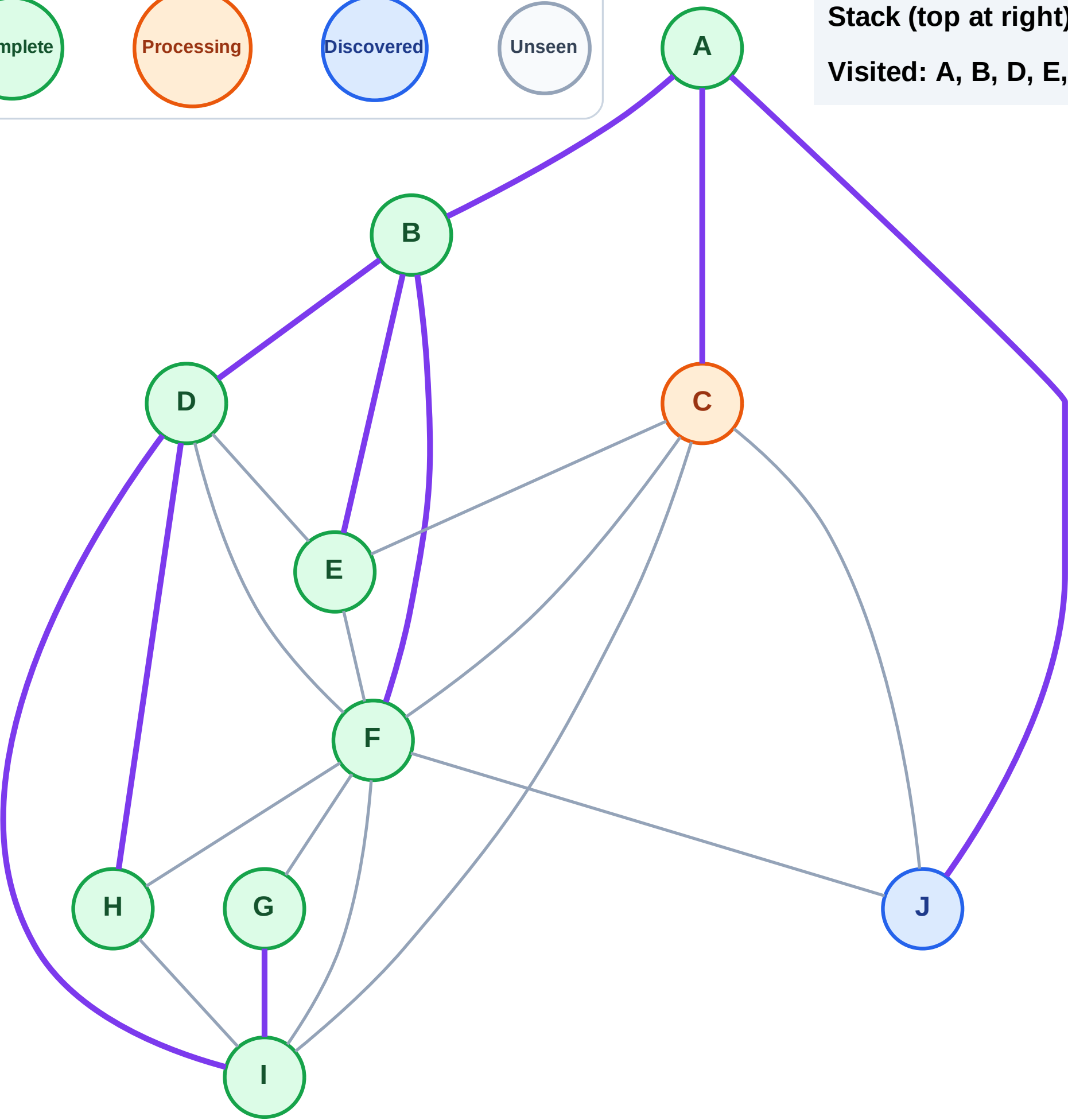
Step 18: Process node C

Legend



Stack (top at right): J

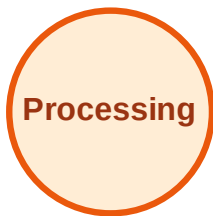
Visited: A, B, D, E, F, G, H, I



DFS Traversal

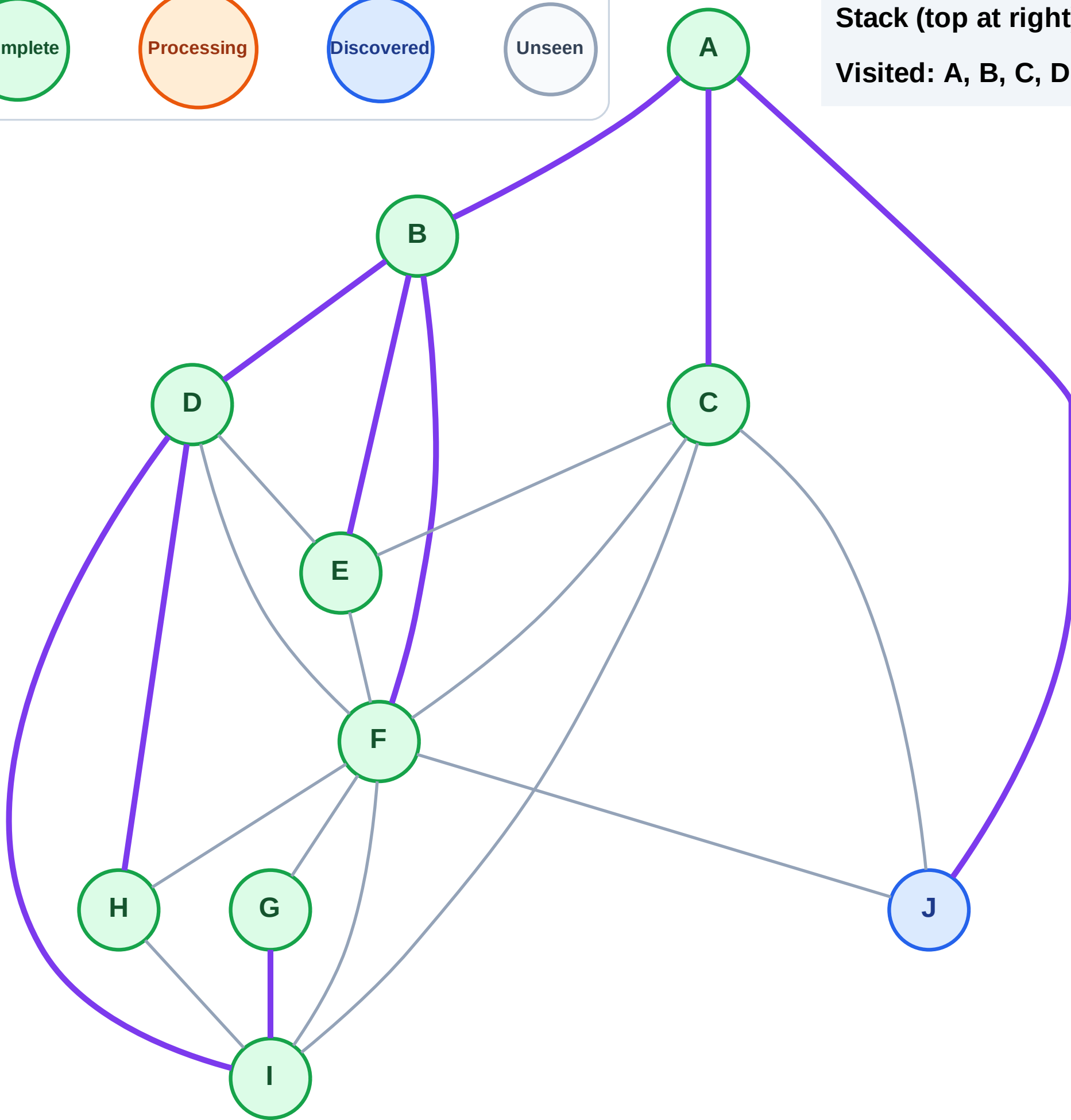
Step 19: No new neighbors from C

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



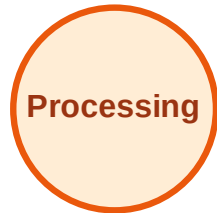
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



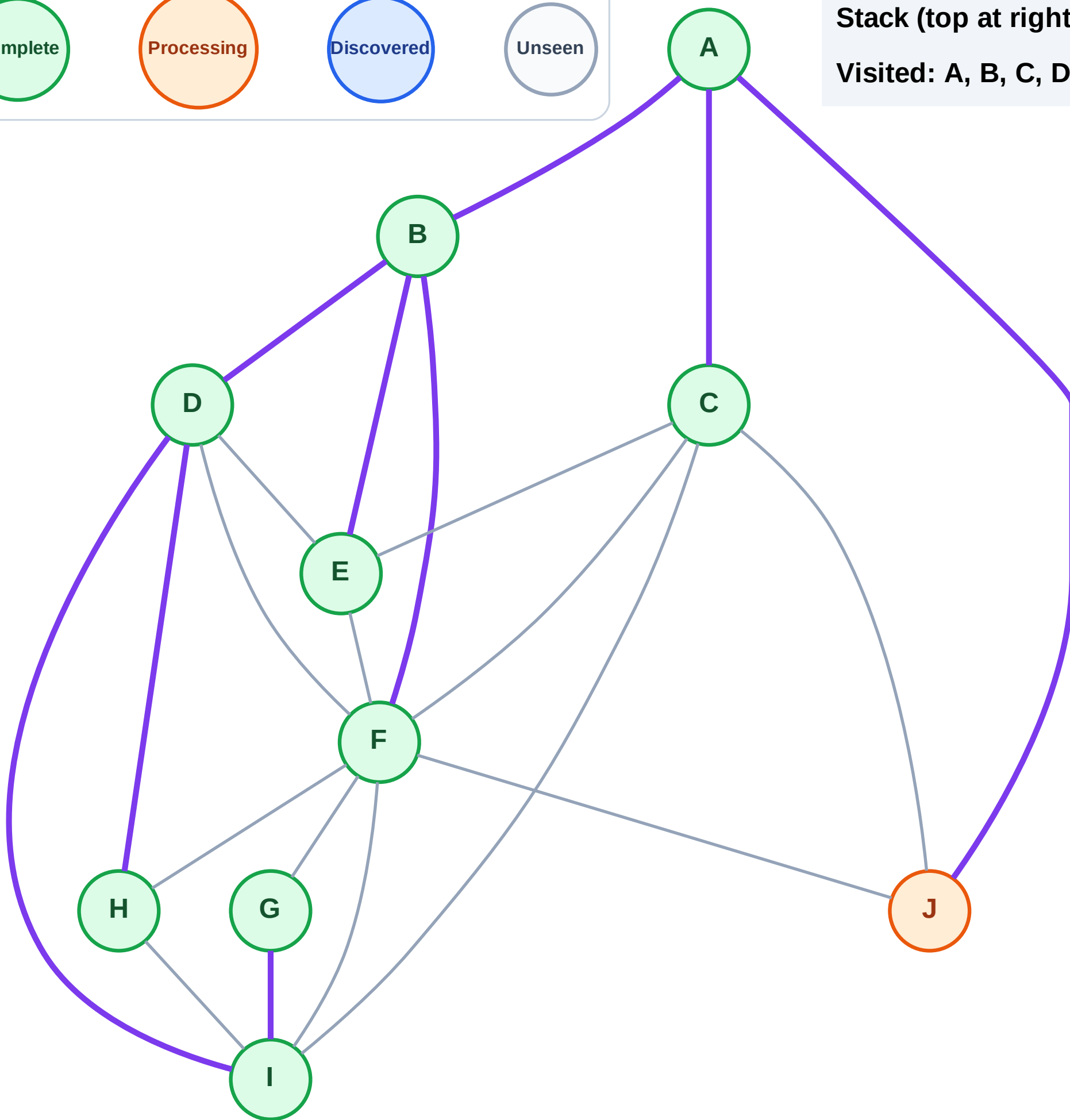
Discovered



Unseen

Stack (top at right): (empty)

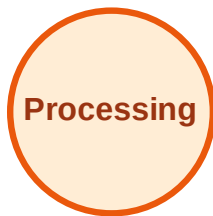
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

